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Modern methods for transgene delivery and genome editing in plants

Diploma thesis

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1. Review the current state of knowledge in the fields of plant genetic transformation and genome editing.
2. Design and assembly of vector constructs for transgenesis of hop (*Humulus lupulus* L.).
3. Assessing the efficiency of genome editing nucleases in cells of selected hop cultivars.
4. Generation of transgenic hop material.
5. Testing the developmental regulators for optimization of hop *in vitro* regeneration.

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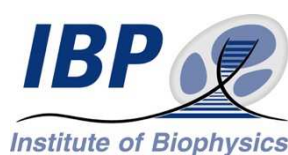
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ABSTRACT

Modern methods for transgene delivery and genome editing in plants

Transgenesis and genome editing are key techniques in the field of molecular genetic manipulation, allowing the editing of genetic information to modify important plant traits. The *Agrobacterium*-mediated transformation is nowadays a sufficiently efficient tool for transgene delivery into the plant genome and was chosen among other methods for our experiment. Nevertheless, many problems can be encountered in this research such as the efficiency of plant cell transformation rate and the ability of *in vitro* regeneration of plant material especially after transgenesis. This work aimed to produce transgenic material. *Humulus lupulus* L. variety 'Saaz' was selected as a model plant for transgenesis. Binary vector constructs for transformation carrying the *FT* and *LFY* gene expression cassettes were designed, assembled, and subsequently tested and a defined level of transformation efficiency was assessed by employing the histochemical GUS test. Also, growth regulators were tested to optimize hop regeneration *in vitro* and the best medium variant for successful regeneration of hop explants (MS-B5 + zeatin + IAA) was observed. The *SOC1* and *JOINTLESS (SVP-like)* genes were also analyzed, and gRNAs for the targeted knockout by using CRISPR/Cas9 were designed. The results of this work show that transgenesis in hop contributes to useful insights into research on this crop. It can be expected that the research will continue to investigate other aspects in the future like introducing and evaluating the effectiveness of genome editing tools and testing alternative genetic constructs on the hop.

Keywords: transgenesis, genome editing, *Agrobacterium*-mediated transformation, developmental regulators, CRISPR/Cas9, vector construct, hop.

ABSTRAKT

Moderní metody přenosu transgenů a editace genomu u rostlin

Transgenoze a editace genomu jsou klíčové techniky v oblasti molekulárně genetických manipulací, které umožňují upravovat genetickou informaci za účelem modifikace důležitých vlastností rostlin. Technika transformace zprostředkovaná bakterií *Agrobacterium* je v současné době dostatečně účinným nástrojem pro přenos transgenů do genomu rostlin a byla vybrána pro experimenty v této diplomové práci. Zároveň existují překážky pro účinnou tvorbu transgenních rostlin, jako je účinnost bakteriální transformace a schopnost regenerace rostlinného materiálu *in vitro*. Cílem této práce bylo získat transgenní materiál. Jako modelová rostlina pro transgenozi byl vybrán chmel otáčivý (*Humulus lupulus* L.) tradiční odrůda Žatecký poloraný červenák (Saaz). Byly navrženy, sestrojeny a následně testovány binární vektorové konstrukty pro transformaci chmele nesoucí kazety pro expresi genů *FT* a *LFY* a byla testována úroveň účinnosti transformace pomocí histochemického testu GUS. Rovněž byly testovány růstové regulátory pro optimalizaci regenerace chmele *in vitro* a byla pozorována nejlepší varianta média pro úspěšnou regeneraci explantátů chmele (MS-B5 + zeatin + IAA). Byly také analyzovány sekvence genů *SOC1* a *JOINTLESS (SVP-like)* a navrženy gRNA pro cílenou mutagenezi CRISPR/Cas9. Výsledky této práce ukazují, že transgenoze u chmele přispívá k získání užitečných poznatků ve výzkumu této plodiny. Lze očekávat, že výzkum touto prací nekončí a v budoucnu bude pokračovat ve zkoumání dalších aspektů, jako je zavádění a hodnocení účinnosti nástrojů pro editaci genomu a testování alternativních genetických konstruktů na chmelu.

Klíčová slova: transgenoze, editace genomu, transformace zprostředkovaná *Agrobacteriem*, vývojové regulátory, CRISPR/Cas9, vektorový konstrukt, chmel.

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1 INTRODUCTION

In an era marked by a growing global population and increasing environmental challenges, ensuring sustainable agriculture is imperative. As global agriculture struggles with the need to produce more food with fewer resources, plant biotechnology emerges as a critical tool in addressing these pressing concerns. The ability to introduce beneficial genes into plants and precisely modify their genomes holds the promise of enhanced crop yields, improved resistance to pests and diseases, and the development of novel agricultural solutions. Biotechnology research is helping us to decipher the potential of genetic engineering, ultimately contributing to our understanding of how science can transform the way we nourish our planet.

Through an exploration of genetic transformation techniques, we have witnessed how advancements in *Agrobacterium*-mediated transformation, biolistic gene delivery, and other innovative methods have enabled the precise introduction of foreign genes into plant genomes.

Despite the rapid development of the field of genome editing, there are technical and regulatory obstacles to its widespread application in agriculture. One of the major challenges is the delivery of nucleases for genome editing into plant cells. To overcome this obstacle, scientists have developed new strategies.

To fully exploit the benefits of genome editing in agriculture, not only technical but also legal and regulatory issues need to be addressed. If these aspects are resolved, it could help to start a real revolution in the agricultural industry, similar to what happened soon after the rediscovery of Mendel's laws.

In this thesis, we decided to select genes suitable for transgenesis, namely *FT* and *LFY*, which we want to overexpress to obtain hop with early flowering. We also want to test a way to inactivate the *SOC1* and *SVP* genes that block the transition to flowering. Judging by the functions and regulatory properties of these genes in hop development, both of these pathways would lead to the desired phenotype.

Our research seeks to harness the potential of genetic engineering to solve pressing problems in agriculture. By modifying the hop genome, we want to pave the way for the development of hop varieties with improved characteristics, which will ultimately contribute to the development of agricultural practices.

2 AIM OF THESIS

This diploma thesis explores cutting-edge techniques in the field of transgene delivery and genome editing in plants. The practical part of this thesis is devoted to the design of vector constructs for transgenesis, evaluation of the efficiency of nucleases for genome editing, creation of transgenic hop (*Humulus lupulus* L.) material, and testing of developmental regulators to optimize hop regeneration *in vitro*.

3 LITERATURE REVIEW

3.1 Transgenesis in Plants

Genetic engineering is a part of biotechnology, which includes a set of approaches and methods that enable changes in the hereditary information of organisms. The use of genetic engineering allows us to create transgenic plants that have improved traits such as resistance to herbicides, pathogens, abiotic stresses, better biochemical composition and taste, and rapid maturation (Tripathi et al., 2023). The cultivation of transgenic crops brings increased yields and early maturity due to their ecological sustainability and plasticity, which brings high economic effects. The cultivation of transgenic crops also contributes to a significant reduction in the utilization of pesticides and mineral fertilizers, characterizing their farming practices as more environmentally friendly than traditional ones (Sood & Lal, 2023).

The incorporation of recombinant DNA technology into plant genetic engineering was based on three outstanding achievements in the field of plant physiology: the discovery of plant growth regulators, the development of methods for culturing plant cells and tissues *in vitro* on specific media, and the determination of the phenomenon of somatic plant cell totipotency. These achievements have paved the way for manipulating plant genomes and creating transgenic organisms with desired traits.

3.1.1 Methods of Transgene Delivery

Transgenic plants have a historical background extending over four decades. The main method of obtaining transgenic plants used to be the direct treatment of meristems, seeds, and pollen with foreign DNA and subsequent screening of the offspring for the presence of introduced genes. Genetic transformation of plants was carried out by microinjecting DNA and electroporation of protoplasts. The integration of diverse gene transfer methods, and tissue culture techniques has facilitated efficient and widespread transformation of various crop plants, enabling the development of transgenics as a valuable tool for targeted single-gene breeding (Gosal et al., 2018; Singh et al., 2022).

3.1.1.1 *Agrobacterium-Mediated Transformation*

The method is based on the naturally occurring gene transfer of plant cells by soil phytopathogenic bacteria *Agrobacterium tumefaciens* or *Agrobacterium rhizogenes*. Such bacterial strains can deliver a foreign gene into the plant genome (Pratiwi & Surya, 2020). Transformed plant cells can be regenerated into normal fertile plants (Fig. 1), in most cases through the stage of callus formation or somatic embryos with subsequent regeneration of plants from them (Gelvin, 2003). These techniques are largely genotype-dependent.

The popularity of this method is due to the relative simplicity of the transformation and the high yield of transgenic plants. As a starting material, to have a strain of *Agrobacterium* harboring a vector construct. The vector must have the sequence of the gene to be introduced into the plant genome. In this case, the transgene should be under the control of a regulatory region (e.g. promoter), ensuring the correct expression in a plant cell. In addition to the functional gene, the vector must have a selectable marker (Ziemienowicz, 2001).

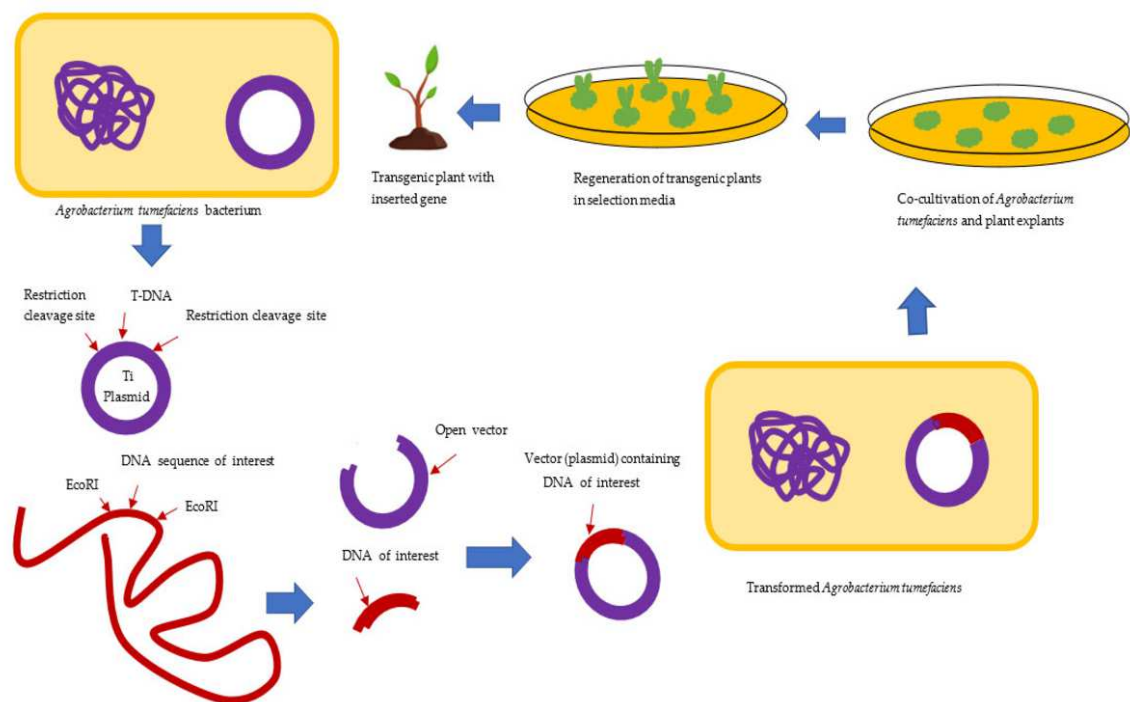


Fig. 1 *Agrobacterium-mediated genetic transformation of the plant. Agrobacterium transformation involves Agrobacterium tumefaciens transferring a segment of its DNA, known as T-DNA, into a plant host's genome. This process relies on virulence genes within Agrobacterium, T-DNA processing, transfer via a specialized secretion system, integration into the plant genome, and subsequent gene expression (retrieved from Ghimire et al., 2023)*

Sterile explants such as leaf discs, young roots, hypocotyls, cotyledons, and internodes are used for transformation. When using formed meristems for *Agrobacterium* transformation, it is possible to obtain transgenic plants regardless of genotype, since plant regeneration from meristems is the simplest method compared to their regeneration from callus or somatic embryos (Chen et al., 2020). Explants are inoculated by a liquid medium containing an *Agrobacterium* with a vector construct, the inoculation time is selected individually for each plant species. In this case, only the wounded surface of the explant is infected. After a few days of cocultivation, the explants are transferred to a selective medium, which leads to the death of non-transformed cells. A few weeks later, the transformed explants develop shoots that are tested for transgene presence and subsequently transferred to the rooting medium.

Moreover, the *Agrobacterium*-mediated plant transformation method is favored for its lower likelihood of unintended epigenetic effects, such as gene silencing and variable expression, in contrast to other plant biotechnology methods, like direct gene transfer and particle bombardment, which often result in multiple DNA insertions and increased risk of epigenetic variations (Ziemienowicz, 2010).

3.1.1.2 Biolistic Gene Delivery

Biolistic gene delivery is a method of delivering genes into plant cells using microprojectiles such as tungsten or gold nanoparticles (Fig. 2). The method works by shooting these microparticles coated with DNA into the plant cells, allowing the DNA to penetrate the cell wall and enter the nucleus (Lacroix & Citovsky, 2020). This method is useful for plant biology research because it allows for the introduction of foreign genes into plant cells without the need for a bacterial vector. Biolistic gene delivery has been used to study gene expression, protein subcellular localization, plant virology, silencing, and targeted genome editing via transient expression of customized endonucleases (Baltes et al., 2017).

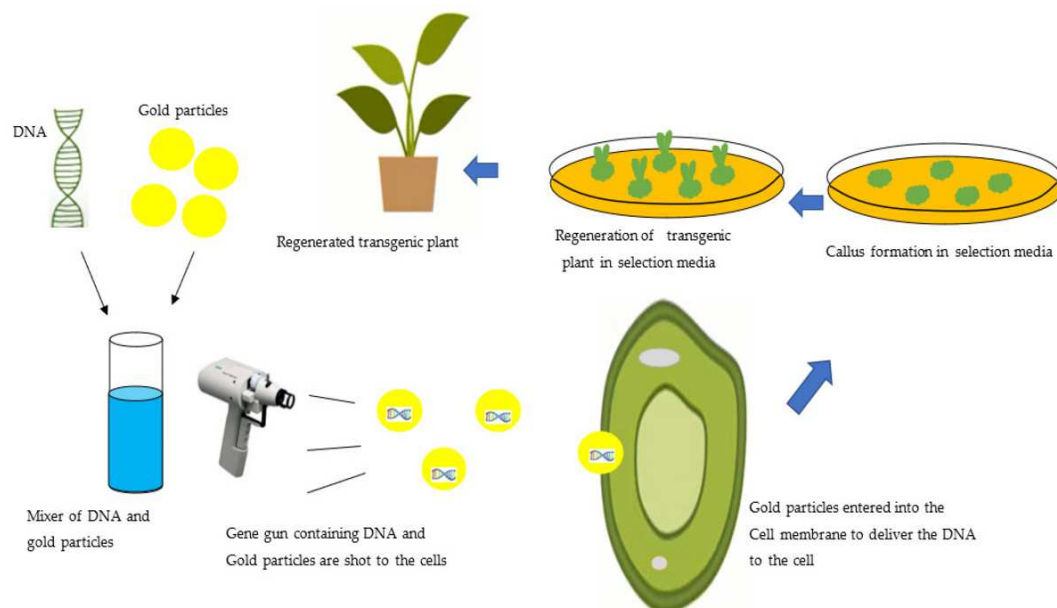


Fig. 2 Biolistics method of genetic transformation of the plant. The method begins with the coating of small, dense particles, often gold or tungsten, with DNA containing the desired genes. These particles are then loaded into a chamber of a device known as a gene gun. Upon activation, the gene gun propels the DNA-coated particles at high velocity toward plant tissues. The impact of the particles on the plant cells results in their penetration into the cell walls and, in some cases, integration into the plant genome. Following bombardment, the transformed plant tissues are cultured in a suitable medium under controlled conditions to promote the growth and development of transgenic cells. Finally, the regenerated plant cells are selected and cultivated to generate whole transgenic plants expressing the desired traits (retrieved from Ghimire et al., 2023).

Notably applicable to both dicot and monocot plants, this approach has demonstrated success in transforming intact plant tissues like leaves and petals. It has been particularly effective in generating transgenic varieties of maize, soybeans, oats, rice, wheat, and barley. One of the key advantages of this method is its time efficiency, as it allows for direct gene transformation of cells and tissues, ensuring higher stability of the resulting transformants. Additionally, its versatility makes it applicable to a wide range of plant species (Ghimire et al., 2023).

3.1.1.3 PEG-Mediated Transformation of Protoplasts

The polyethylene glycol PEG-based transformation method is applied to plant cell protoplasts from which the cell wall has been removed (Díaz & Koop, 2014). In this method, the protoplasts are placed together with PEG vesicles containing plasmid DNA. This allows the protoplasts to engulf and integrate the foreign DNA into the plant genome (Fig. 3). Although the PEG method requires enzymatic tissue cleavage to isolate

protoplasts, it is a more economical option because it does not depend on a specialized and expensive delivery system (Yu et al., 2020).

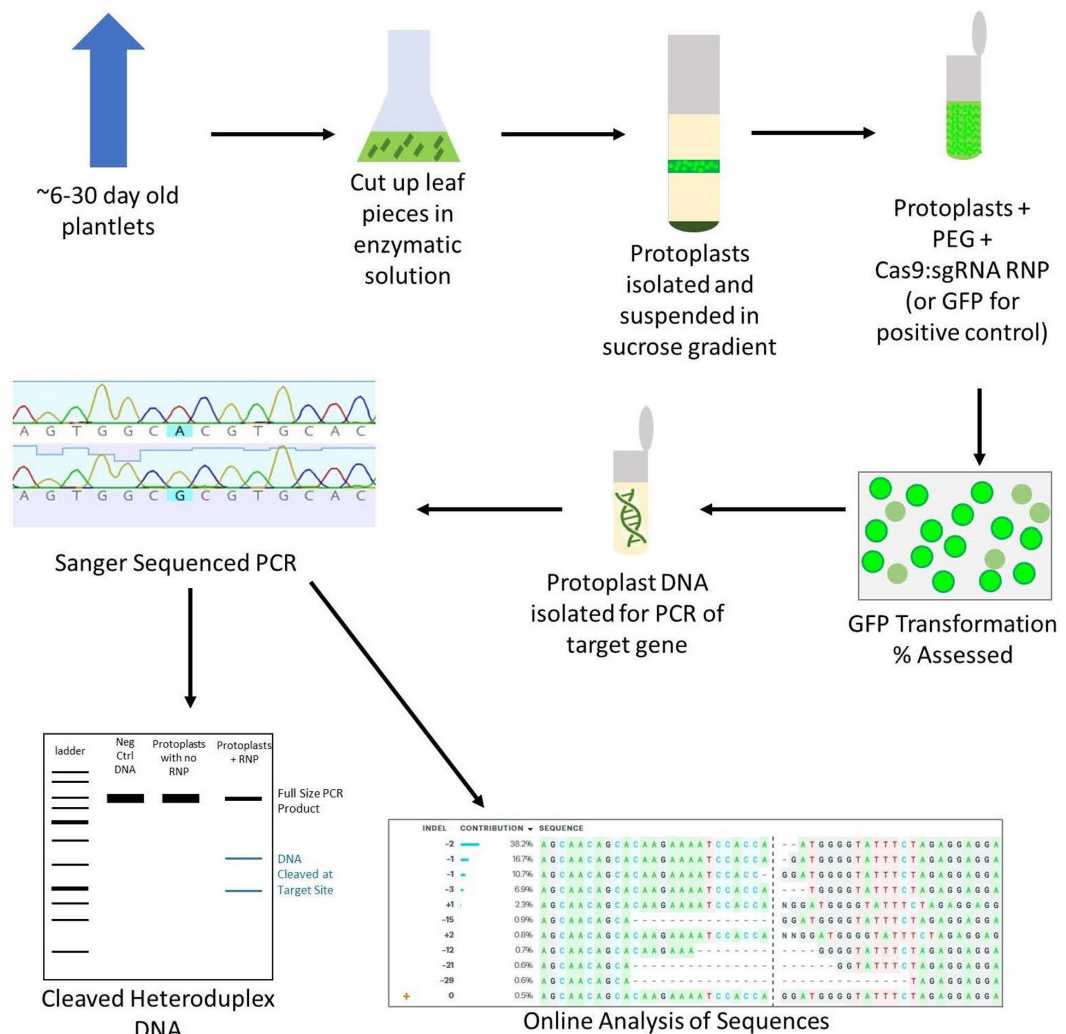


Fig. 3 Summary of the protocol for isolating and transforming protoplasts. The protocol for isolating and transforming protoplasts involves several sequential steps. Initially, plant tissues are collected and treated with cell wall-digesting enzymes to release protoplasts. The resulting protoplast suspension is then purified through filtration or centrifugation to remove debris and cell wall remnants. Next, the isolated protoplasts are transformed with foreign DNA using various methods (e.g. PEG-mediated transformation). Following transformation, the protoplasts are incubated under specific conditions to allow for DNA uptake and integration into the genome. The success of the transformation is verified by sequencing followed by further bioinformatic analyses (retrieved from Brandt et al., 2020)

Transformation of plant protoplasts is used for direct gene transfer into plant cells. The *neomycin phosphotransferase* gene is used as a marker gene that determines resistance to kanamycin and some other aminoglycoside antibiotics (Fuchs et al., 1993).

This method has several disadvantages, such as the complexity of the procedure for obtaining protoplasts, the inefficient *in vitro* regeneration from protoplasts for many plant

species, the possibility of somaclonal variants, and a long period from the moment of protoplast isolation to the production of transgenic plants (Shao et al., 2023).

Another protoplast transformation method is based on the fact that high-pressure pulses reversibly increase the permeability of biomembranes (Hapala, 1997). For plant protoplasts, the electroporation procedure has proven to be very effective. The method consists of the following: plant protoplasts, which are in a high-concentration solution containing DNA vectors, are exposed to a high-voltage pulse. As a result, DNA molecules are absorbed by the cells through the pores in the cell membrane. After the solution is diluted, the protoplasts are seeded on an appropriate medium for regeneration. The efficiency of the transfer is determined 24-48 hours after the electric shock (Bates, 1995).

The CRISPR/Cas9 system and TALEN can induce targeted double-stranded DNA breaks in the genome, facilitating homologous recombination for targeted gene insertion. High expression of Cas9 may enhance homologous recombination efficiency in transgenic plants (Yu et al., 2020). In addition, overexpression of morphogenic genes may increase regeneration efficiency in tissue culture, potentially facilitating plastid transformation (Yu et al., 2020). Protoplast regeneration, previously considered labor-intensive, has become more feasible with improved tissue culture protocols and laboratory equipment.

The PEG-mediated protoplast transformation has shown promise in a variety of cultures, offering an efficient alternative to biolistics for the production of transgenic plants. With further advances in technology and methodology, protoplast transformation may become more accessible and efficient for various crop species (Hayashimoto et al., 1990; Wu et al., 2020).

3.1.1.4 Microinjection Method

Compared to other transformation methods, the microinjection method has several advantages: versatility and standardization of operations; a wide range of recipient objects required for transformation (isolated protoplasts, single callus cells, cells within the whole plant, multicellular structures); different molecular genetic material (DNA molecules, isolated nuclei, chloroplasts and mitochondria) used for injection; high transformation efficiency; the possibility of using isolated microspores and pollen cells as objects (Neuhaus & Spangenberg, 1990). The disadvantages include the availability of rather complicated equipment.

This method makes it possible to solve several plant transformation tasks that cannot be solved by other methods, for example – the transformation of plant genotypes for which isolation, cultivation of protoplasts, and regeneration of transgenic plants from them have not been developed and which are not susceptible to inoculation with *Agrobacterium*; transfer of isolated cell organelles into cells; use of non-selective DNA molecules for transformation; use of haploid cells - microspores, from which regeneration of haploid plants is possible in the future (Su et al., 2023).

3.1.1.5 Liposome Encapsulation

This is one of the methods used to protect exogenous material introduced into plant protoplasts from nucleases that destroy nucleic acids. Liposomes are spherical vesicles whose membranes consist of phospholipids. They can be obtained as a result of sharp shaking or sonication of aqueous phospholipid emulsions. Liposomes have been used to introduce *Tobacco mosaic virus* (TMV) RNA, T-plasmid DNA, and whole metaphase chromosomes into plant protoplasts (Fraley et al., 1982). The advantages of the liposome transfer system include low toxicity to plant cells and the possibility of using them for most plant species whose cells are capable of liposome utilization (Gad et al., 1990). At the present stage, this method is rarely used due to its technical complexity and low transforming activity.

3.1.2 Preparation of Genetic Constructs for Transgenesis

The preparation of bacterial vector constructs for transgenesis involves a careful and systematic approach using molecular biology techniques. The whole process begins with the identification and isolation of the gene of interest and then its amplification using polymerase chain reaction (PCR) or other molecular cloning techniques (Nora et al., 2018). At the same time, appropriate regulatory elements, such as promoters and terminators, are selected or designed to control the expression of the gene in the desired manner (Nora et al., 2018). Subsequently, these genetic elements are assembled into the vector backbone using molecular cloning techniques, thus ensuring precise orientation and sequence fidelity.

The resulting vector construct undergoes thorough validation by sequencing and functional assays to confirm its integrity and functionality before transgenesis.

In some cases, the process of introducing a genetic construct into a plant may be inefficient or unpredictable. This can be caused by many factors, such as cell stress during transformation, the presence of growth inhibitors, or the limited ability of plant cells to accept and integrate external genetic materials. Also, some genetic constructs may be unstable or cause undesirable phenotypic effects in plants. Therefore, carefully test engineered transgenic lines for stability and desirable traits before further use.

3.1.2.1 Selection of the Promoter

The selection of a promoter is a very important issue especially in cases to express an introduced gene in specific tissues or individual organs, at certain stages of ontogeny, under the influence of any compounds (Villao-Uzho et al., 2023). It was found that some promoters are complex structures consisting of subdomains that control gene expression. Thus, for the most studied 35S promoter of *Cauliflower mosaic virus* (CMV), deletion analysis showed that when certain fragments of the promoter sequences were deleted, the transcription efficiency decreased dramatically and was only a small part of the original level (Odell et al., 1985).

3.1.2.2 Selectable Marker Genes

Selectable marker genes provide plant cells with the ability to survive exposure to otherwise toxic substances, such as antibiotics or herbicides, thus allowing for the identification of transformed cells/tissues. Commonly used selectable markers include *nptII*, *hpt*, *bar*, and *gox*, which provide resistance to kanamycin, hygromycin, phosphinothricin, and glyphosate, respectively (Mehrotra & Goyal, 2016; Hamzeh et al., 2015). These marker genes encode proteins that grant resistance to specific toxic compounds, enabling the selection of transformed cells that have successfully integrated the foreign DNA. While the *nptII* gene, providing resistance to kanamycin, is widely utilized in dicotyledonous plants, alternative positive selection systems, such as isopentenyl transferase and phosphomannose-isomerase, have been developed to avoid the use of antibiotic or herbicide resistance genes, aiming for improved cotransformation efficiency in *Agrobacterium*-mediated transformation methods (Mehrotra & Goyal, 2016).

Recent progress in agricultural biotechnology has introduced chemically inducible site-specific recombinase systems, which efficiently regulate the removal of selectable

marker genes from crop genomes when they are no longer needed (Hare & Chua, 2002). These advancements address concerns about genetic modification and aim to enhance public acceptance of agricultural biotechnology.

3.1.3 Regeneration of Transgenic Plants

The selection and regeneration of transgenic plants represent crucial steps in plant biotechnology, facilitating the identification and propagation of genetically modified organisms. Following selection, transgenic plants undergo a regeneration phase, during which transformed cells proliferate and develop into whole plants (Vidal et al., 2010; Singh et al., 2023). This regeneration process can be achieved through tissue culture techniques, such as organogenesis or somatic embryogenesis, depending on the plant species and genotype. The successful regeneration of transgenic plants is critical for the establishment of stable transgenic lines and subsequent functional analysis (Thangadurai et al., 2020).

3.1.4 Molecular Analysis of Transgenic Plants

Molecular analysis of transgenic plants constitutes a fundamental aspect of plant biotechnology, enabling the characterization and validation of genetic modifications introduced into plant genomes. This analysis involves a diverse array of techniques aimed at assessing the presence, expression, and stability of transgenes, as well as evaluating potential off-target effects and unintended genomic alterations (Thangadurai et al., 2020). Polymerase chain reaction (PCR) assays are commonly employed to confirm the presence of transgene inserts, while quantitative PCR (qPCR) allows for the precise quantification of transgene expression levels. Additionally, Southern blot analysis provides insights into transgene integration patterns and copy number variations within plant genomes (Trijatmiko et al., 2016). Furthermore, transcriptomic approaches, such as RNA sequencing (RNA-Seq), enable comprehensive profiling of gene expression changes associated with transgene insertion or manipulation. Genome-wide techniques, including whole-genome sequencing and bioinformatic analyses, are utilized to identify potential off-target effects and unintended mutations.

3.1.5 Phenotypic Assessment

The phenotypic assessment of transgenic plants is a critical aspect of plant biotechnology, allowing to evaluation of the effects of genetic modifications on plant morphology, physiology, and performance. This assessment involves the comprehensive analysis of various phenotypic traits, including but not limited to growth characteristics, yield parameters, stress tolerance, and nutritional content.

Morphological assessments encompass measurements of plant height, leaf size, root architecture, and flowering time, providing insights into the overall development and architecture of transgenic plants. Physiological assessments, such as photosynthetic efficiency, water use efficiency, and nutrient uptake, offer valuable information on the functional attributes of transgenic plants under different environmental conditions.

Additionally, stress tolerance assays, including drought, salinity, and disease resistance tests, elucidate the impact of genetic modifications on plant resilience and adaptation to adverse conditions. Furthermore, biochemical analyses enable the quantification of metabolites, vitamins, and secondary compounds in transgenic plants, revealing alterations in nutritional quality and bioactive content (Cheeran et al., 2017).

The integration of these phenotypic assessments facilitates the comprehensive characterization of transgenic plants and informs breeding strategies for the development of improved crop varieties.

3.2 Functional Genomics

Genomics, proteomics, and bioinformatics have become the main direction of life science development in the 21st century. The acceleration of the formation of these science disciplines was greatly facilitated by the implementation of the Human Genome Project. This program united the efforts of various scientific teams around the world and created conditions for solving many problems of modern biology not only at the molecular but also at all levels of the existence of living matter. Nowadays, genomics has become the main engine of life science development. It is divided into the following sections, which are united by a common ideology created by modern genetics, as well as methods, approaches, and technologies (Cullis, 2003 | Functional Genomics).

Genomics created conditions for the emergence and development of such sciences as proteomics and bioinformatics, as well as modern biotechnology, i.e. technologies based on obtaining products using methods of cell and molecular biology and genetic engineering (Cullis, 2003 | Bioinformatics). It should be noted that modern genomics is a relatively young discipline and there are ongoing discussions regarding its structure and interrelationships with other branches of science. However, the structure of modern genomics can be schematically represented as follows (Fig. 4).

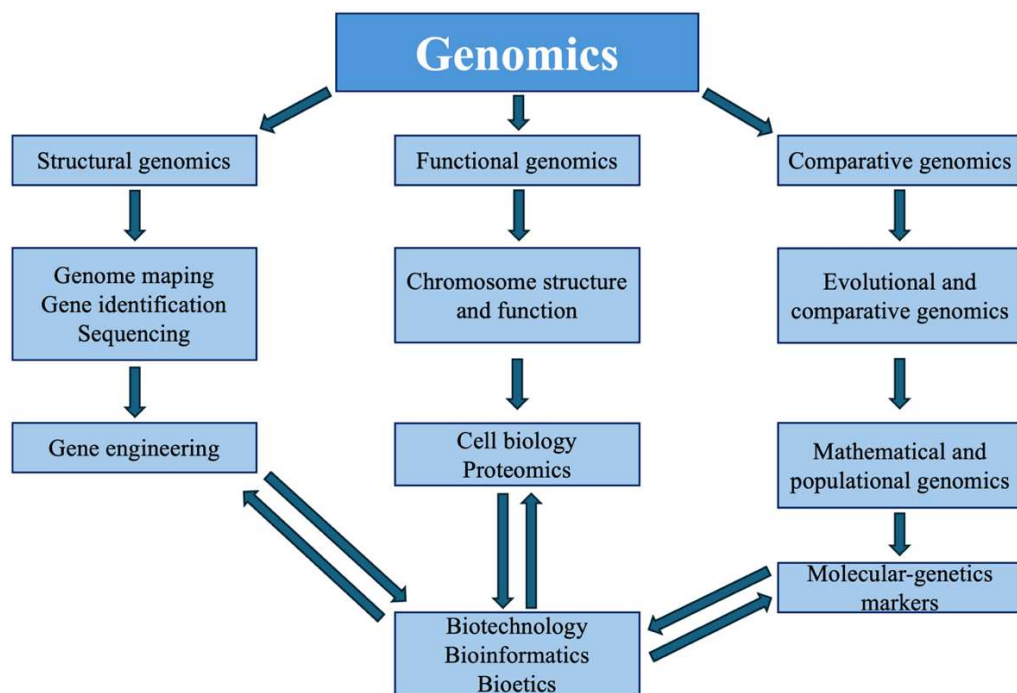


Fig. 4 Structure of modern genomics (inspired by Cullis, 2003 | Functional Genomics)

The study of the plant genome lags far behind the study of the human genome in terms of its pace and costs. Plant genomes in some species are several times larger than the human genome. The genomes of major crop plants, especially rice, are not as large. *Arabidopsis*, a small dicotyledonous plant with a very short growing season and a genome size of 125 million bp, and rice, whose genome has 420-470 million bp, were chosen as model objects (The *Arabidopsis* Genome Initiative, 2000; Kurata et al., 2002). Currently, work is underway to study the genome of other plants.

3.2.1 Methods of Functional Genomics

New methods for detecting working genes using microarrays (which allow simultaneous detection of the activity of thousands of genes), as well as recent advances in selective spectroscopy (targeting specific components or properties of a sample while disregarding others), quickly became significant methods for functional plant genomics. In general, it should be noted that modern genomics integrates knowledge from other disciplines, not only molecular genetics, but also cytology, cell biology, and biotechnology (Mannstadt & Wein, 2018). An equally important and promising branch of genomics is genetic engineering and transgenesis.

The development of genomics has led to the emergence of technologies for sequencing, genome annotation, genome editing, and gene expression analysis. Sequencing methods such as Illumina, Pacific Biosciences, and Oxford Nanopore Technologies facilitate the obtaining of huge amounts of sequence data. After genome sequencing, bioinformatics methods are used to annotate and predict gene function. Genome editing technologies, such as CRISPR/Cas9, TALEN, and ZFN, enable specific genome alterations (Gaj et al, 2013). Gene expression analysis methods such as microarrays and RNA-Seq allow gene expression levels to be measured with high accuracy and sensitivity.

In conclusion, the integration of these cutting-edge genomic technologies not only enhances our understanding of plant biology but also paves the way for innovative approaches to address agricultural challenges.

3.2.1.1 Microarray Analysis

DNA microarrays are a key element of modern functional genomics, which, thanks to miniaturization and automation, makes it possible to obtain whole genome information

from multiple samples (Galbraith, 2006). Microarrays can be synthesized either *in situ* (oligonucleotide microarrays) or deposited on a solid surface (cDNA microarrays). They are used to monitor gene expression and analyze DNA variability, e.g. to detect mutations and polymorphisms. Microarray analysis is a high-throughput method that is used to measure the expression of thousands of genes simultaneously (Girke et al., 2000).

Unique DNA sequences (oligonucleotides or cDNA) are fixed on a glass slide at microscopic spots (the probe). RNA is extracted from control and treated samples and labeled with different colors of fluorescent dye (Meilan, 2004). The labeled RNA is hybridized with a complementary single-stranded DNA probe, after which the microarray is scanned with a laser (Meilan, 2004). The fluorescent images are superpositioned, and the color of each dot indicates the difference in gene expression in the samples.

In agricultural and breeding research, microchips are applied to study various aspects of the plant life cycle, e.g. circadian rhythms, defense mechanisms against external influences, stress responses, fruit ripening processes, and phytochrome signaling pathways and seed development (Aharoni & Vorst, 2002).

3.2.1.2 RNA Sequencing

RNA sequencing, or RNA-Seq, is a powerful tool for transcriptome analysis (Wang et al., 2009). It provides quantitative information about gene expression levels, alternative splicing events, and RNA modifications. In general, RNA-Seq begins with the isolation of total RNA from cells or tissues, followed by cDNA synthesis, library preparation, and sequencing (Fig. 5). The resulting data can be analyzed to identify novel transcripts, quantify gene expression levels, and investigate transcriptomic changes in response to different biological conditions.

There is also a more precise version of this technique that allows sequencing of the genetic material of individual cells. Single-cell RNA sequencing (scRNA-Seq) is a cutting-edge technology used to analyze the transcriptome of individual cells (Fig. 5) (Wang et al., 2023; Bawa et al., 2022). Unlike traditional bulk RNA sequencing, which provides an average expression profile of a population of cells, scRNA-Seq allows researchers to study the gene expression patterns of individual cells within a heterogeneous population (Bawa et al., 2022). This technique provides insights into cellular heterogeneity, cell-to-cell variability, and rare cell populations that may be masked in bulk analysis.

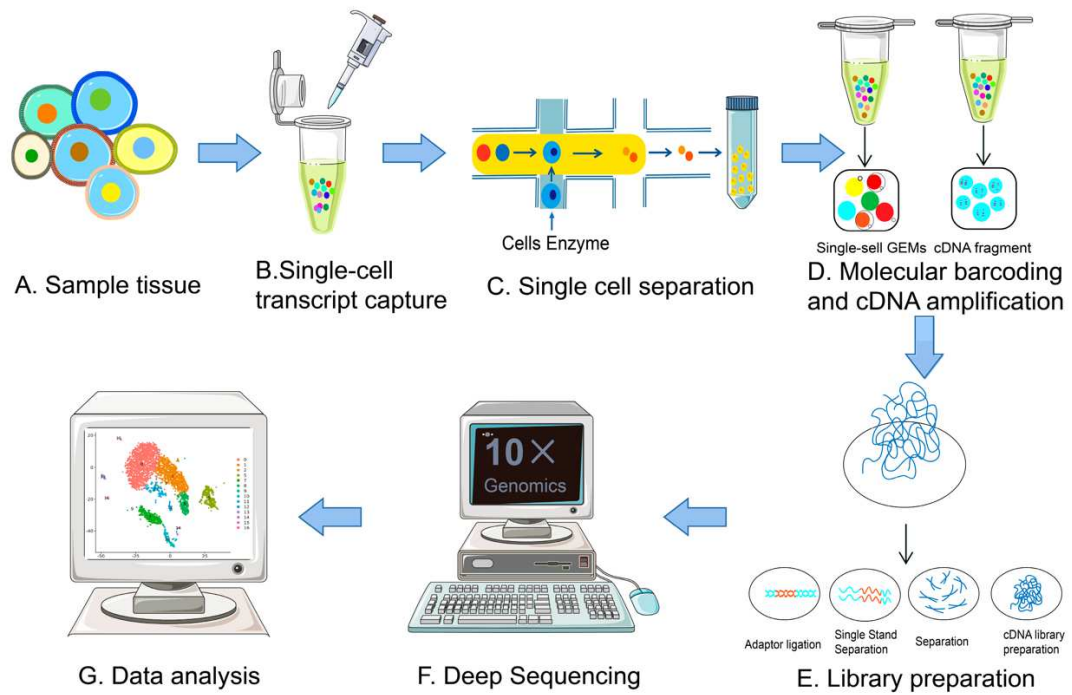


Fig. 5 Scheme of the RNA-Seq protocol. The single-cell RNA sequencing (scRNA-Seq) protocol involves several key steps, including cell isolation, cell lysis, RNA capture, reverse transcription, cDNA amplification, library preparation, and sequencing, enabling the analysis of gene expression profiles at the single-cell level (retrieved from Wang et al., 2023)

RNA sequencing techniques such as RNA-Seq and single-cell RNA sequencing revolutionize our understanding of gene expression dynamics at both bulk and single-cell levels, offering invaluable insights into molecular mechanisms underlying biological processes.

3.2.1.3 Chromatin Immunoprecipitation Sequencing

Chromatin Immunoprecipitation Sequencing (ChIP-Seq) is a technique used to investigate protein-DNA interactions and chromatin modifications at a genome-wide level. It involves the immunoprecipitation of protein-DNA complexes using antibodies specific to the protein of interest or even a particular post-translation modification, followed by sequencing of the associated DNA fragments (Fig. 6) (Mannstadt & Wein, 2018).

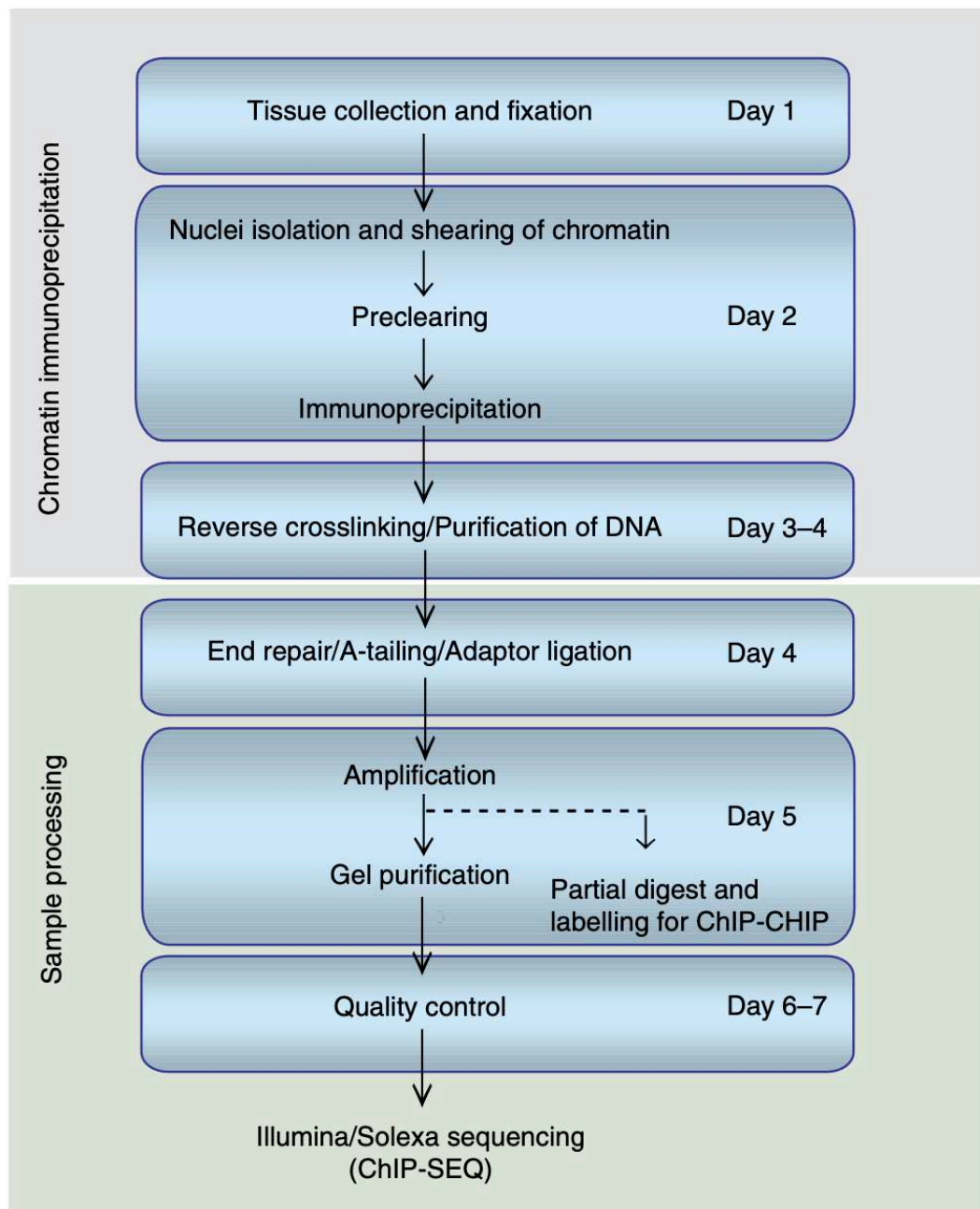


Fig. 6 Different steps in a typical ChIP experiment and sample processing. In a typical Chromatin Immunoprecipitation (ChIP) experiment, the steps involve crosslinking of DNA and proteins, cell lysis and chromatin shearing, immunoprecipitation using antibodies against the protein of interest, washing to remove the nonspecific binding, elution of the DNA-protein complexes, reversal of crosslinks, and finally DNA purification and analysis, allowing for the identification of DNA sequences associated with the target protein (retrieved from Kaufmann et al., 2010)

Using this method in plant research is hampered by problems such as the availability of specific antibodies or the need to genetically modify plants to express labeled proteins, which might be labor-intensive, time-consuming, and expensive. Improved ChIP-Seq methods have been developed to address these problems. These utilize clones of labeled

transcription factors transiently expressed in protoplasts, providing a more cost-effective and time-efficient approach for studying transcription factor binding sites (TFBS) landscapes e.g. in *Arabidopsis* and other plants (Wang et al., 2021).

3.2.1.4 Functional Proteomics

Functional proteomics encompasses a variety of techniques aimed at studying the structure, function, and interactions of proteins on a global scale. This includes methods such as mass spectrometry-based proteomics, protein-protein interaction assays, and protein localization studies. By characterizing the proteome of cells or tissues under different conditions, functional proteomics provides valuable information about protein function, signaling pathways, and cellular responses to stimuli (Vanderschuren et al., 2013). Integration of proteomic data with other omics datasets enhances our understanding of gene regulation and biological processes at the protein level.

Mass spectrometry-based proteomics provides valuable insights into plant physiology, allowing large-scale analysis of plant proteins (Yan et al., 2022). Bottom-up proteomics studies provide extensive lists of peptides, which improves the characterization of protein quantities and their post-translational modifications (PTMs). Methods such as data-dependent acquisition (DDA) and data-independent acquisition (DIA) improve proteome coverage and accuracy (Dakup et al., 2023). Methods such as metal oxide affinity chromatography (MOAC) enrich the coverage of the phosphoproteome and help us to study dynamic phosphorylation events (Wolschin & Weckwerth., 2005). Organelle proteomics provides a comprehensive view of protein localization and function in plant cells, facilitating the study of organelle dynamics and their relationship to metabolites.

Proteomics plays a key role in studying plant responses to abiotic stresses such as high and low temperatures, drought, and salinity (Sha Valli Khan et al., 2014). Under low temperatures, proteins associated with energy metabolism increase, while defense-related proteins decrease. Exposure to high temperatures leads to the induction of heat shock proteins (HSPs), which are crucial for plant adaptation to stress (Meyer & Chartier., 1983; Sha Valli Khan et al., 2014). Studies of tobacco leaf mesophyll protoplasts have revealed the synthesis of specific HSPs under heat stress (Meyer & Chartier., 1983). Also, proteomics has greatly expanded our understanding of plant development and growth by

identifying key proteins involved in different developmental stages such as cell differentiation, organ formation, and flowering (Yan et al., 2022). Analyzing the proteome during plant development provides insight into the molecular events that determine growth and morphogenesis, which is important for improving yield and agricultural practices (Vanderschuren et al., 2013).

Proteomic analysis helps to identify proteins that are critical for processes such as photosynthesis and nutrient uptake, allowing breeders to improve crop productivity by selecting or developing high-yielding varieties (Le et al, 2016). In addition, proteomics helps to improve crop resistance to pests, diseases, and environmental stresses by identifying proteins associated with defense mechanisms and stress response (Vanderschuren et al., 2013).

3.2.1.5 CRISPR/Cas9 Genome Editing

Recent advances in the sequencing of plant genomes have opened up great opportunities to study their functions and responses to external influences. Thanks to this increase in genomic data, we can better understand the structure of genomes and how they change in response to environmental conditions (Atkins & Voytas, 2020). The development of new methods of gene editing expands the possibility of precise intervention in genetic processes, which allows us to customize plant properties with greater accuracy (Hudzieczek et al., 2022).

Initial efforts in plant genome editing utilized zinc finger nucleases (ZFNs), which incorporate the DNA-binding domain of a zinc finger and the DNA cleavage domain from the FokI type IIS restriction endonuclease (Gaj et al., 2013). While ZFNs have been successfully used for mutagenesis in *Arabidopsis*, soybean, tobacco, and maize, their complex design has made widespread use difficult (Arora & Narula, 2017). The TALENs (transcription activator-like effector nucleases) technology is based on prokaryotic proteins that target plant promoters to activate genes after bacterial infection. TALENs allow for precise targeting of specific genomic sequences, which ensures their successful application for genome editing in different cultures. However, the biggest breakthrough in plant genome editing came with the invention of CRISPR/Cas9 nucleases, which are based on the bacterial immune system and require only a 20-nucleotide gRNA sequence to target the specific genomic region (D'orso et al., 2023). CRISPR/Cas9 has greatly

simplified the development and implementation of genome editing by allowing multiplex gene editing in plants (Hudzieczek et al., 2022).

CRISPR/Cas9 technology has revolutionized functional genomics by enabling precise genome editing in a wide range of organisms (Doudna & Charpentier, 2014). It utilizes a guide RNA (gRNA) to target the Cas9 nuclease to specific genomic loci, where it induces double-strand breaks (Fig. 7). These interruptions can undergo repair through error-prone non-homologous end joining (NHEJ) or homology-directed repair (HDR) mechanisms, facilitating gene knockout, insertion, or regulation of gene expression (Fig. 7) (D'orso et al., 2023). CRISPR/Cas9-based screens enable systematic investigation of gene function on a genome-wide scale, facilitating targeted mutagenesis of genes involved in various biological processes (Asmamaw & Zawdie, 2021).

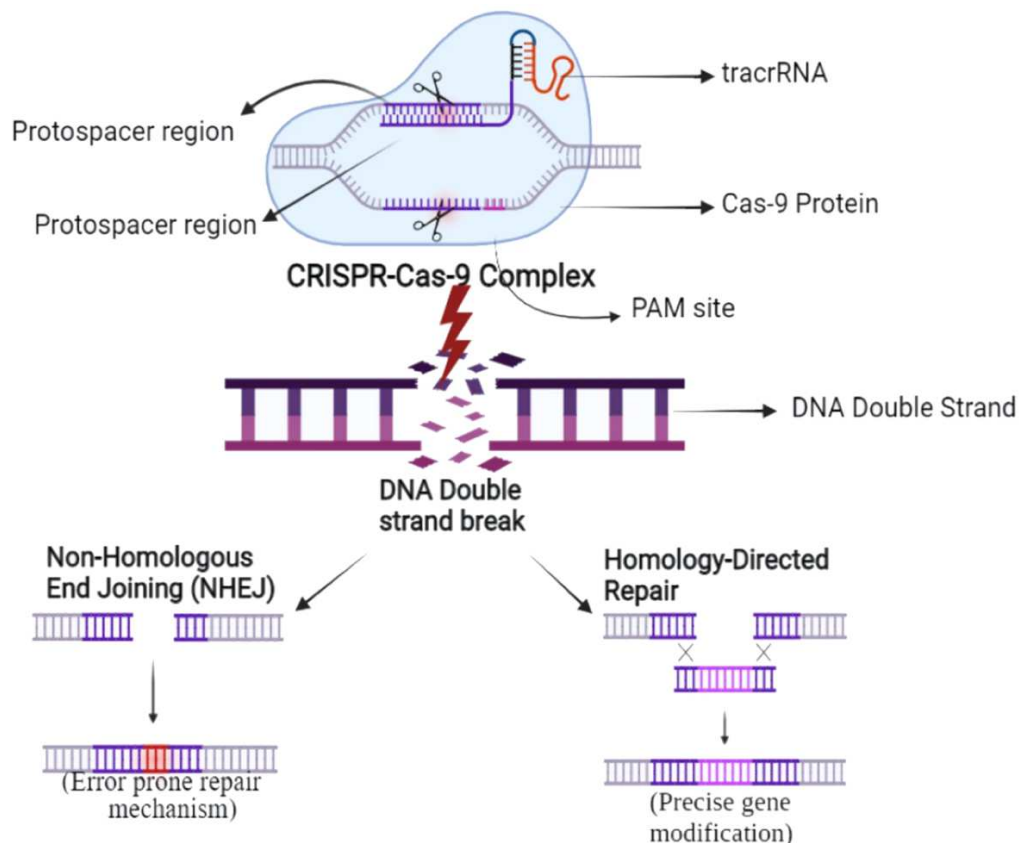


Fig. 7 CRISPR/Cas9-mediated genome engineering method. The CRISPR/Cas9-mediated method involves designing a guide RNA (gRNA) to target a specific DNA sequence, forming a complex with the Cas9 enzyme. Upon locating the target site, Cas9 creates a double-strand break (DSB) in the DNA. Subsequently, two repair mechanisms, non-homologous end joining (NHEJ) and homology-directed repair (HDR), come into play. NHEJ directly ligates the broken DNA ends, often resulting in indels and gene disruption, while HDR uses a homologous DNA template to precisely repair the DSB, allowing for gene knockout, knock-in, or correction of mutations (retrieved from Javaid et al., 2022)

CRISPR tools are also used to target gene expression in plants using RNA-guided transcriptional activators and repressors derived from dCas9 (a modified version of the Cas9 enzyme with a mutation in the active site of the endonuclease that disables its ability to cut DNA but preserves its ability to bind to target DNA sequences). Various strategies are used to activate genes using CRISPR, including CRISPRa, which allows the recruitment of protein activators to the target promoter region. There are also CRISPR tools for targeted transcriptional repression, such as CRISPR interference. In addition, CRISPR/Cas13 RNA endonucleases can induce post-transcriptional RNA silencing of endogenous genes and immunity to RNA viruses in plants (Sharma et al., 2022). CRISPR-based genome editing expands broad prospects for precise modification of genetic sequences in plant biology, making it an important breeding tool (Hudzieczek et al., 2022).

3.3 Application of Genetic Engineering Techniques in Agriculture

The formal date of birth of genetic engineering is considered to be 1972, when Paul Berg's group in the United States created the first *in vitro* DNA recombinant, which combined genetic material from three sources: the complete genome of the oncogenic monkey *Simian virus* (SV40), part of the genome of the temperate bacteriophage K and the genes of the galactose operon of *E. coli* (Jackson et al., 1972).

Plants have been genetically engineered for various purposes, including scientific research, ornamental purposes, and agricultural enhancements. Techniques such as tissue culture and genetic modification have been particularly successful in plants like tobacco and *Arabidopsis thaliana*, which serve as model organisms for genetic engineering studies (Atkins & Voytas, 2020; Arora & Narula, 2017). In basic research, plants are modified to understand gene functions by removing or altering specific genes and observing resulting phenotypic changes.

The leading biotechnology companies Syngenta, Monsanto, DuPont, and Bayer spend more than two billion US dollars annually on research. First-generation GMO plants exhibit resistance to total herbicides, insects, and viruses, along with modified fatty acid composition or hybrid seed production systems (e.g. chicory, papaya, pumpkin, and tomatoes).

In general, in the European Union GMO plants are particularly negatively perceived (Hundleby & Harwood, 2019). Although the area under their cultivation is relatively small, the data on the creation of a new generation of GMO plants by companies, and universities in recent years is striking.

Modern genome editing techniques have ushered in a new era for agriculture, offering powerful tools to precisely modify the phenotype of crops (Atkins & Voytas, 2020). This chapter explores the diverse applications of genome editing in agriculture (Fig. 8), emphasizing the potential for enhancing crop traits, improving yields, and addressing global food security challenges.

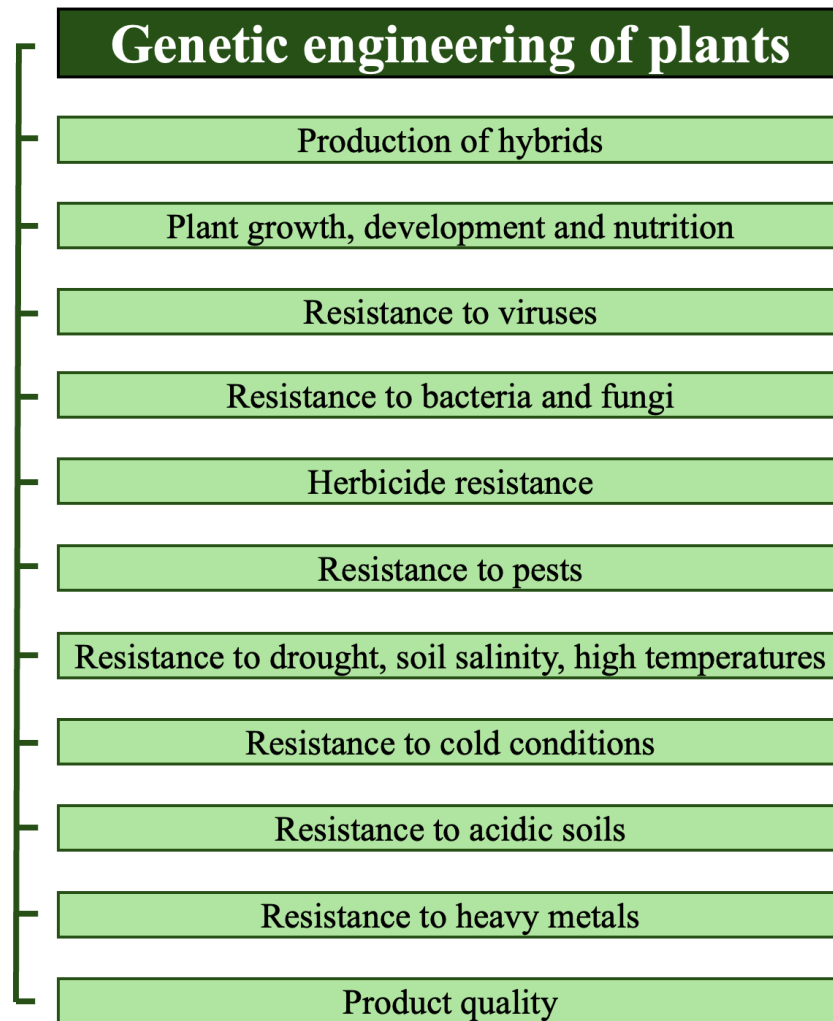


Fig. 8 *The main tasks of plant genetic engineering*

With the advent of genetic modification of plants in the 70s and 80s, scientists were able to localize genes and control their functions, opening the way for the transfer of genes between organisms to give them new properties.

The commercial use of genetically modified foods began in 1994, starting with GM tomatoes and GM corn, which had certain useful characteristics, such as resistance to diseases, pests, or herbicides (Xu, 2022). For example, Monsanto's GM potatoes have a built-in gene encoding a toxin that kills the Colorado potato beetle, making the growing process easier (Kadoić Balaško et al., 2020). China also grows GM cotton that produces the insecticide Bt toxin, which protects the plant from pests (Abbas, 2018). In the United States and Canada, GM soybeans, which are resistant to the Roundup herbicide, are widespread (Benbrook, 2016; Abbas, 2018). Later generations of genetically modified crops aimed to enhance nutritional quality and offer non-food applications like pharmaceutical production, biofuel generation, and bioremediation.

3.3.1 Herbicide Resistance

The widespread introduction of genetically modified (GM) herbicide-tolerant crops has raised concerns about the development of herbicide-resistant weeds. While the switch to glyphosate, a common herbicide, initially showed an improvement in environmental impact factors, it has led to the emergence of glyphosate-resistant weeds, especially in countries such as the United States, Brazil, and Canada (Benbrook, 2016). However, research shows that this problem is not only due to the nature of GM crops but rather to the fact that glyphosate was originally used as the only method of weed control.

In response to this problem, GM crop producers have been advised to use other herbicides with different mechanisms of action alongside glyphosate, as well as to adopt agricultural practices such as minimum tillage systems to control weeds more effectively (Xu, 2022). It is also important to note that herbicide resistance is not unique to GM crops, as herbicide-resistant weeds have been emerging in conventional agriculture over time. Even before the widespread introduction of GM technologies, herbicide-resistant weeds were already present, which suggests that this problem predates the use of GM crops by several decades (Bawa & Anilakumar, 2013).

Despite the progress made by genetically modified (GM) crops, they face hurdles, notably a reduction in research funding for novel herbicides. Initially, the efficacy of glyphosate resulted in decreased exploration of alternative herbicides and a decline in investment toward discovering new herbicidal agents. As a consequence, weed populations are evolving resistance to glyphosate, posing a challenge to current crop management strategies. While new herbicide-resistant (HR) crop technologies provide temporary fixes, achieving sustainable results requires a holistic approach that includes varied strategies for managing plants and the discovery of herbicides with unique properties (Green & Owen, 2011).

Growers should incorporate both selective and HR crop-enabled herbicides into their practices. They need to anticipate changes in weed populations and implement measures to prevent the dominance of challenging-to-control and HR weeds. Employ balanced strategies that integrate chemical, mechanical, and cultural methods to optimize weed management effectiveness. In essence, while HR crops have transformed weed control, achieving sustainability demands a comprehensive approach that extends beyond reliance on a single herbicide. It is crucial to diversify tactics and engage in ongoing research to combat the emergence of resistant weeds.

3.3.2 Virus Disease Resistance

Plant viruses are completely dependent on the host to replicate and spread, and whether they cause disease depends on the interaction between the plant's defense mechanisms and the virus itself. Plants have innate defenses and evolved strategies against viruses. Natural defenses include innate immunity, RNA silencing, translational repression, autophagy, and inhibition of viral movement (Green & Owen, 2011). Engineered resistance includes methods such as pathogen resistance and gene editing, particularly through CRISPR/Cas technology (Majumdar et al., 2023). Breeding programs comprising transgenesis and/or gene editing offer the hope of creating virus-resistant plants by incorporating various resistance genes.

Genetic modification has also been proposed for conserving endangered plant species by making them resistant to invasive species and diseases. For example, transgenic papaya plants were developed to resist the *Papaya ringspot virus* (PRSV) (Tecson Mendoza et al., 2008).

3.3.3 Pest Resistance

Transgenic plants resistant to insects and pests are a large group. The bacterium *Bacillus thuringiensis* produces a protein (delta-endotoxin) that is toxic to many insect species. The gene for this protein is transferred into the genome of plants that are not damaged by insects. In this way, insect-resistant soybeans, corn, cotton, and potatoes have been created (Abbas, 2018).

The creation of insect-resistant transgenic forms, called Bt plants, has an environmental impact in addition to its commercial value and prospects (Tabashnik et al., 2023). It is well known that when insecticides are applied to plants, only a small percentage of the insecticides reach the surface and act effectively, and a huge portion of applied insecticide leaches into the environment. They end up in water, soil, and ultimately edible plant products, and they also kill beneficial insects.

The development of insect-resistant plants will lead to a significant reduction in the use of insecticides, and the restoration of beneficial insect populations and will have a positive impact on flora, fauna, and the biosphere (Tabashnik et al., 2023).

3.3.4 Abiotic Stress Tolerance

One of the main areas of plant genetic engineering comprises the production of varieties that are resistant to various abiotic stressors such as extreme temperatures, water deficit, excessive moisture, excessive light, lack of light, soil salinity, toxic metals (e.g. soil pollution), and other environmental factors that can adversely affect plant growth and development.

For instance, transgenic rice plants can express stress-inducible *OsDREB2A*, resulting in improved resistance to dehydration stress and heightened productivity (Cui et al., 2011; Todaka et al., 2015). Similarly, in the case of salinity tolerance, plantain plants were genetically modified to express a *maize cystatin* gene and a synthetic peptide, rendering them more resistant to nematode infestations and better able to withstand salinity stress (Parmar et al., 2017). Moreover, *Arabidopsis* plants engineered with *C-repeat binding factor (CBF)* genes demonstrated increased resilience to freezing temperatures by regulating the expression of cold-responsive genes (Jaglo et al., 1998; Abdullah et al., 2022; Edrisi et al., 2023). Furthermore, *Brassica juncea* plants expressing metallothionein genes showed improved tolerance to heavy metals like cadmium and zinc by sequestering toxic metals (Małecka et al., 2021; Zhigang et al., 2006; Pan et al., 2018).

3.3.5 Improved Nutritional Content

Transgenic plants represent a promising avenue for addressing nutritional deficiencies in crops by enhancing their content of essential amino acids and other vital nutrients.

One well-known example of genetically modified crops aimed at enhancing nutrient value is Golden Rice, which produces beta-carotene, a precursor of vitamin A. Golden Rice addresses dietary deficiencies prevalent in regions with limited access to vitamin A-rich foods, potentially reducing child mortality and blindness caused by vitamin A deficiency (Kobayashi et al., 2023). Soybeans have been genetically modified to increase methionine content, thereby improving their overall nutritional quality (Guo et al., 2020). Similarly, genetic engineering has been employed to elevate lysine and tryptophan levels in maize, making it a more comprehensive source of protein (Le et al., 2016). In *Arabidopsis*, the expression of a bacterial dihydrodipicolinate synthase transgene has significantly boosted lysine content in seeds (Zhu & Galili, 2003). Additionally, transgenic tomatoes have been developed to accumulate the antioxidant

pigment astaxanthin, which can benefit fish health when incorporated into aquaculture diets (Huang et al., 2013; Enfissi et al., 2019). Beyond these specific examples, genetic modification has also been utilized to enhance the levels of antioxidants like carotenoids and flavonoids in various crops (Zhu et al., 2013).

Moreover, ongoing research aims to improve the nutritional profile of staple crops through genetic engineering, promising further advancements in agricultural nutrition and food security.

3.4 Environmental Impact and Regulatory Landscape

Genetically modified plants have great potential for improving yields and protecting crops, but they also raise outrage and controversy over their safety and environmental impact. Indeed, this is a big problem. Genetically modified organisms, if used uncontrollably, can cause significant damage. To prevent this, strict biosafety measures must be followed.

Ethical and regulatory considerations play a pivotal role in the development and deployment of transgenic plants, ensuring the responsible conduct of research and the safe deployment of genetically modified organisms into the environment. Ethical considerations encompass a broad range of concerns, including potential ecological impacts, biodiversity preservation, and socio-economic implications. Researchers must weigh the potential benefits of transgenic plants, such as increased crop yields and enhanced nutritional content, against potential risks, such as unintended environmental consequences or socio-economic disparities (Karalis et al., 2020). Additionally, ethical considerations extend to issues of intellectual property rights, equitable access to technology, and transparency in scientific communication.

Government oversight of genetically modified organisms (GMOs) varies between countries and involves regulatory frameworks established over time. Beginning in 1975, guidelines were set at the Asilomar meeting, and later, the Cartagena Protocol on Biosafety in 2000 provided international regulations for the transfer, handling, and use of GMOs (Beeckman & Rüdelsheim, 2020). Each country has its regulatory bodies overseeing GMO research and release, with universities and research institutes usually needing approval for genetic engineering experiments.

There are differences in GMO regulation between regions, notably between the United States and Europe. The European Union distinguishes between approval for cultivation and import/processing of GMOs, with fewer GMOs approved for cultivation but many for import/processing (Hundleby & Harwood, 2019). Some countries ban or restrict GMOs, while others permit them with varying degrees of regulation. Illegal releases have occurred due to enforcement weaknesses.

Despite the serious complications associated with biosafety requirements, the biotech business is steadily increasing worldwide. The total area of GMO plants in the world reaches 190.4 million hectares and continues to expand (ISAAA, 2019). The European Commission advocates for compulsory disclosure and traceability to provide consumers

with information and facilitate the withdrawal of products if any adverse effects emerge. Conversely, groups such as the American Medical Association argue that disclosure in the absence of scientific evidence of harm could be misleading (Genetically modified organism, 2024 | *Wikipedia*; Bruetschy, 2019). Disclosure policies vary globally, with some nations requiring disclosure for products surpassing a specific GM content threshold, while others permit voluntary disclosure. The rising popularity of products identified as non-GMO indicates a growing consumer desire for transparency regarding GMOs (Wunderlich & Gatto, 2015).

Regulatory agencies typically require comprehensive risk assessments, including environmental impact assessments, allergenicity testing, and molecular characterization of transgenic plants, prior to their approval for commercial use (Weale, 2010). Moreover, labeling requirements and traceability systems are often implemented to facilitate consumer choice and mitigate potential risks.

The likelihood of negative consequences from the use of genetically modified organisms in practice is low. In the natural environment, no serious environmental problems of transgenic plant varieties and hybrids have been identified (Bauer Panskus et al., 2020). However, their potential danger is still quite controversial (Fig. 9) (Hundleby & Harwood, 2019; Bawa & Anilakumar, 2013).

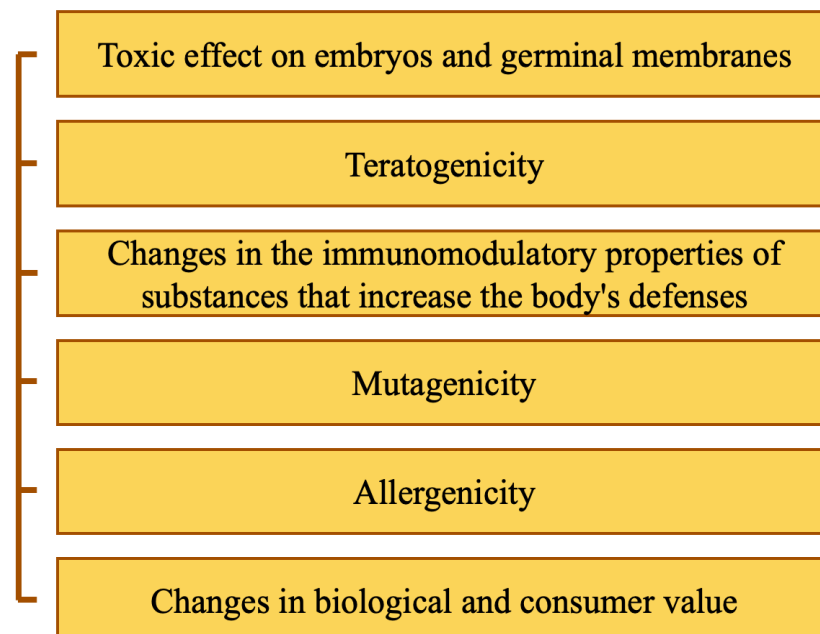


Fig. 9 *Potential food hazards of using transgenic crops*

Many transgenic crops are resistant to insects, produce proteins that can block digestive enzymes not only in insects but also in humans, and affect the pancreas (Dona & Arvanitoyannis, 2009). Transgenic corn and tomatoes that are resistant to insects and pests can produce substances that break down into toxic and mutagenic compounds, which in turn can pose a direct threat to humans, such as an increased risk of cancer and mutations. Some genetically modified plants are resistant to insects and pests due to their increased lectin content, but they can be mutagenic and have a significant impact on human embryos (Bawa & Anilakumar, 2013). As a result of intracellular processes, some genetically modified rice varieties accumulate biologically active enzyme degradation products (which help increase yields) that can trigger the formation of malignant tumors (Dona & Arvanitoyannis, 2009). Transgenic sugar beet resistant to glyphosate accumulates a significant amount of its breakdown products, which are potent carcinogens (Bawa & Anilakumar, 2013).

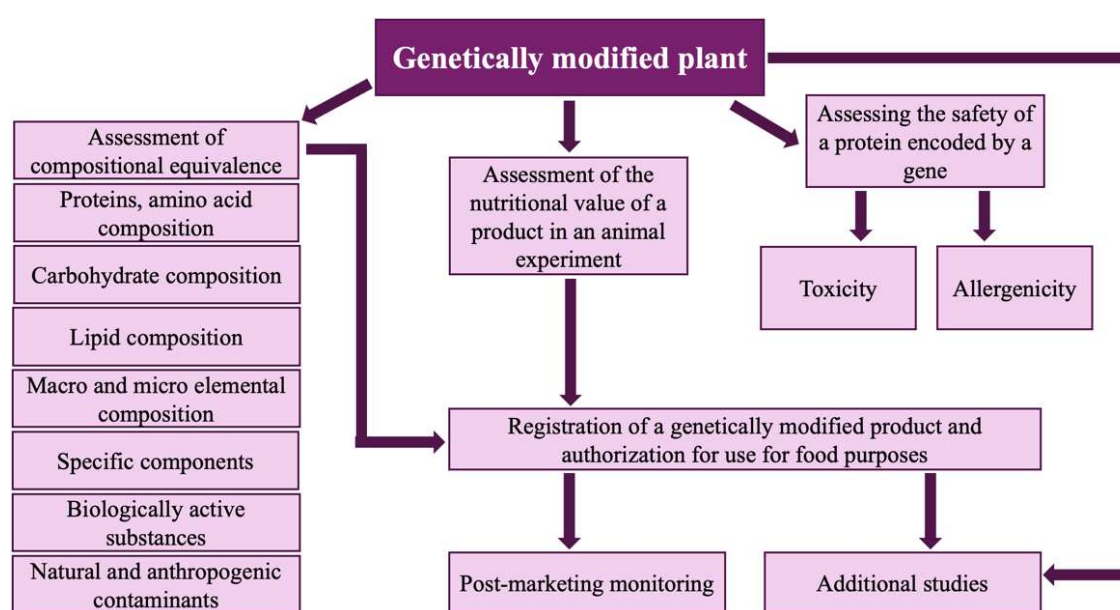


Fig. 10 *System for assessing the quality and safety of genetically modified sources*

Today, the modern concept of compositional equivalence is generally accepted, based on the comparison of genetically modified food sources with their traditional counterparts in terms of phenotype, content of key nutritional and anti-nutritional substances, toxins that are standardized in food, allergens and biologically active components characteristic of this type of product. In addition, the form in which the product is traditionally used for food and how it reacts to technological processing are taken into account. The chemical

composition of the final product and its nutritional and energy value are analyzed. The research program is determined for each specific product individually (Fig. 10).

In conclusion, the integration of ethical principles and robust regulatory oversight is essential to navigate the complex landscape of transgenic plant development and deployment, balancing the potential benefits of biotechnology with societal values and environmental stewardship (Genetically Modified Organisms (GMOs) | Learn Science at Scitable, 2008).

3.5 Genetic Engineering of Hop

Hop (*Humulus lupulus* L.) is a perennial herbaceous plant that belongs to the *Cannabaceae* family. Hop is an important and widely known crop in Czech agricultural production and its cultivation and research have a long tradition (Patzak, 2002). It is used in the brewing industry. Female hop flowers impart bitterness, aroma, and flavor to beer. Genetic engineering holds great promise in enhancing hop plants to meet the demands of the brewing industry, improve agricultural sustainability, and explore new applications.

3.5.1 Description and Characterization of Hop

Hop is a liana with a quadrangular hollow stem, which can reach up to 10 m in length. The plant has a long creeping rhizome.

The leaves of common hop are three- to five-lobed with coarsely serrate margins and intertuberculate bracts. The plant is dioecious, with male inflorescences forming multi-branched panicles and female flowers in cone-shaped compound inflorescences.

Flowering occurs in summer, and the fruits, which are nuts with spirally coiled buds, ripe in August - September. Hop is widely distributed in the temperate climates of Eurasia, North America, South America, and even northern Africa (Rangel et al., 2021). It can be found almost everywhere in the European part, Western Siberia, the Caucasus, and Altai. The plant prefers to grow on rich soils in forests, bushes, near water bodies, and in plantations.

The hop female flowers, known as "cones," are used in medicine to create drugs such as sedatives. Cones contain valuable chemicals including essential oils, hop resins, vitamins, bitter acids, and other compounds. These substances give hop their aroma, bitter taste, and medicinal properties. However, the internal use of hop requires strict dosage due to its toxicity.

3.5.2 Transformation and Targeted Mutagenesis of Hop

Genetic manipulation of hop plants has already been carried out by some scientific groups and has led to successful results in this field.

A highly efficient method for directly regenerating hop plants, particularly the "Tettnanger" cultivar, has been developed, including the successful *Agrobacterium*-mediated genetic transformation. Shoot internodes from *in vitro* cultures were found to be the most effective explant type, with around 50 % showing direct

organogenesis on a specific medium (Horlemann et al., 2003). This method yielded 21 transgenic plants from 1,440 initially transformed explants, achieving a transformation efficiency of 1.5 % (Horlemann et al., 2003).

The groundbreaking use of CRISPR/Cas9 in hop plants marks a significant milestone in plant genome editing. For example, by targeting the endogenous reporter *phytoene desaturase (PDS)* gene, researchers achieved efficient editing, leading to mutations that disrupted carotenoid biosynthesis. Through *Agrobacterium*-mediated transformation, Cas9 and guide RNAs were introduced into hop explants, resulting in successful editing in a third of the transformed plants (Awasthi et al., 2021). Analysis revealed decreased chlorophyll and carotenoid levels in mutant lines, confirming functional disruption of the *HIPDS* gene (Awasthi et al., 2021). This advancement highlights the potential of CRISPR/Cas for precise editing of the hop genome.

Examples of these advances hold promise for improving hop traits such as pathogen resistance while maintaining other desirable characteristics but also point to one of the most important challenges in creating transgenic hop explants - low transformational efficiency.

3.5.3 Selection of Candidate Genes for Transgenesis

Interestingly, the hop genome lacks orthologs of the *FLOWERING LOCUS C (FLC)* gene, a known flowering repressor associated with vernalization (Gutiérrez et al., 2022). This suggests that hop do not require vernalization for flowering transition. Overall, understanding the phenological cycle and flowering regulation in hop is essential for optimizing cultivation practices, especially in non-temperate regions.

The phenological cycle of hop includes rhizome establishment, vegetative growth in spring and summer, and development of reproductive structures in late summer and early fall. Flowering occurs when the plants reach a certain size and is not influenced by vernalization or rhizome age. Notably, hop thrive in photoperiods greater than 13 hours of light and are considered short-day plants.

The use of hop as a model system for studying the genetic basis of biological processes is limited by the length of the annual life cycle. In general, model plants can produce several generations of offspring in a year, but for hop, this is usually only one generation. Accelerated flowering can be induced by overexpression of genes involved in the vegetative to generative transition.

3.5.3.1 *FLOWERING LOCUS T*

The *FLOWERING LOCUS T* (*FT*) gene plays a critical role in initiating the transition from vegetative growth to flowering by promoting the expression of flower identity genes. Together with the *LEAFY* (*LFY*) gene, *FT* helps to regulate the activity of *APETALA1*, a key regulator of flower development. Overexpression of *FT* results in an early flowering phenotype by accelerating the floral transition (Klocko et al., 2016; Kardailsky et al., 1999; Kobayashi et al., 1999).

Unlike temperate regions, hop in subtropical areas flower year-round, primarily depending on internal cues rather than day length. Identified *HIFT3* and *HIFT5* as potential florigens, with their expression peaking during the hop's developmental window between 15 and 20 nodes (Gutiérrez et al., 2023). Promoter analysis revealed differences: *HIFT3* is hormone-responsive, while *HIFT5* is light-responsive. A model was proposed to explain hop floral development, suggesting that *HIFT3* acts as the primary florigen, while the function of *HIFT5* remains uncertain. Moreover, miRNA families related to reproductive transition have been identified, such as the MIR156 and MIR172 families. miR156 expression was found to decrease during development inversely to miR172 and synchronously with the upregulation of *HIFT3* and *HIFT5* (Fig. 11) (Gutiérrez et al., 2023).

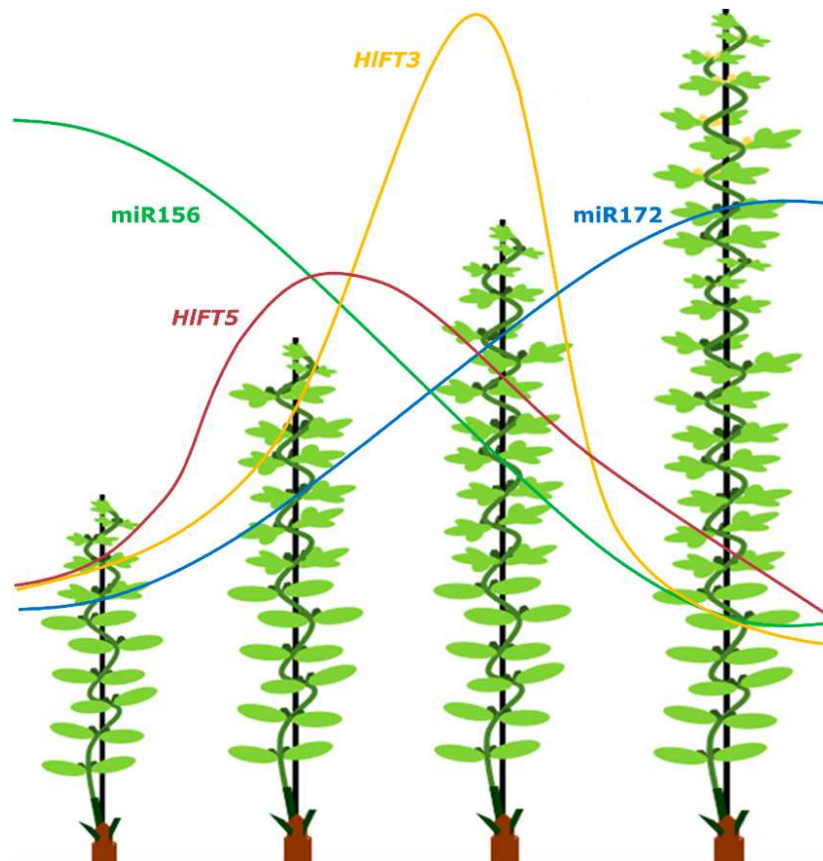


Fig. 11 A scheme of the miRNA156/172 and FT expression explaining the hop floral transition. The curves show the level of expression of the corresponding genes concerning the height of the hop plants (retrieved from Gutiérrez et al., 2023)

3.5.3.2 *LEAFY*

The *LEAFY* (*LFY*) gene is a crucial regulator of floral development in plants, playing a central role in determining when and where flowers are formed. Gene *LFY* encodes a transcription factor essential for determining the identity of the early floral meristem. In cooperation with *APETALA1*, *LFY* facilitates the transition of inflorescence meristems into floral meristems. While overexpression of *LFY* results in earlier flowering, its primary function is to ensure the timely initiation of flower development.

When the *LFY* gene is overexpressed, plants are less sensitive to environmental conditions and flower earlier than expected (Siriwardana & Lamb, 2012). If the gene is mutated and is no longer able to perform its functions, lateral shoots form instead of flowers.

3.5.3.3 SUPPRESSOR OF OVEREXPRESSION OF CO1

The transition from vegetative to reproductive development in plants is a critical phase controlled by floral integrators like *SUPPRESSOR OF OVEREXPRESSION OF CO1* (*SOC1*). *SOC1* gene acts as a transcription activator involved in controlling flowering time. It integrates signals from various pathways, including photoperiod, vernalization, and autonomous floral induction. *SOC1* also regulates the expression of class B and C homeotic genes and mediates the effects of gibberellins on flowering under short-day conditions. Additionally, *SOC1* plays a role in regulating the expression of *LEAFY* (*LFY*), linking floral induction with floral development.

ChIP-Seq analysis revealed that *SOC1* binds to its promotor and to genes involved in flowering time regulation (Immink et al., 2012). MADS domain flowering proteins also bind to *SOC1* regulatory sequences, highlighting the central role of *SOC1* in flowering timing (Immink et al., 2012).

3.5.3.4 SHORT VEGETATIVE PHASE

SHORT VEGETATIVE PHASE (*SVP*) gene acts as a transcriptional repressor, inhibiting floral transition in the autonomous flowering pathway regardless of photoperiod and temperature. Its activity is dosage-dependent. Along with *AGL24* and *API*, *SVP* controls the identity of the floral meristem and regulates the expression of class B, C, and E genes.

Overexpressing a gene similar to *SVP-LIKE* in poplar resulted in a significant delay in the initiation of flowering in three different genotypes of transgenic poplars grown in the field (Goralogia et al., 2021). Increased expression of this gene was associated with a postponed onset of flowering and reduced floral abundance.

3.5.4 Challenges and Considerations

Creating transgenic hop involves introducing foreign genes into hop plants to confer specific traits. Challenges include ensuring stable integration of the transgene into the hop genome for consistent expression across generations and under various environmental conditions. Preventing gene flow to wild or non-transgenic hop varieties is essential for maintaining genetic diversity. Assessing the potential allergenicity or toxicity of the transgenic product is critical for human safety. Public perception of genetically modified crops can influence market acceptance. Additionally, preventing cross-pollination and

stacking multiple transgenic traits present further complexities but offer enhanced benefits such as improved crop yield and disease resistance (Mall et al., 2018).

Apart from its traditional use in brewing beer, hop has gained attention for various alternative applications. Research has highlighted hop potential in phytotherapy, particularly for managing excitability, mood disorders, and sleep problems. Compounds found in hop cones like xanthohumol and isoxanthohumol are being investigated for their potential anti-cancer (Krajnović et al., 2022). Additionally, hop components are being studied for their antimicrobial properties in animal feed, which could drive up demand for this crop (Turner et al., 2011).

Hop breeding for organic systems involves a variety of techniques, including the development of tetraploid and triploid plants to improve desirable traits such as vigor and seedlessness (Trojak-Goluch & Skomra, 2018). The focus is on brewing and agronomic qualities, with consideration given to disease resistance, yield, and brewing characteristics. Traditional breeding has focused on European varieties, but wild American and Asian varieties are gaining attention for their unique traits. Agronomic traits important for organic systems include disease resistance and nutrient use efficiency.

4 MATERIALS AND METHODS

To study the function of genes related to commercially significant traits in hop, previously established the hop genetic transformation procedure. We plan to increase the efficiency of agroinfection and *in vitro* regeneration of transgenic plants using novel genetic engineering approaches. This will allow us to use CRISPR/Cas9 gene editing and other methods of genetic engineering on a large scale. The main objective of this study is to develop hop lines with early flowering, which will greatly improve the efficiency of hop research. In this project, we will combine state-of-the-art genetic engineering methods to provide a tool for functional genomics and the improvement of hop cultivars.

4.1 Materials

4.1.1 Bacterial Material

Escherichia coli TOP10 strain (*mcrA*, Δ (*mrr-hsdRMS-mcrBC*), *Phi80lacZ(del)M15*, Δ *lacX74*, *deoR*, *recA1*, *araD139*, Δ (*ara-leu*)7697, *galU*, *galK*, *rpsL(SmR)*, *endA1*, *nupG*) cells were used for the preparation of vectors for transgenesis at all stages. Then, after the assembly of the binary vector constructs, cells of two different *Agrobacterium tumefaciens* strains, GV3101 (pTiC58DT-DNA) and EHA105 (pTiBo542DT-DNA), were used. Bacterial cultures are stored as 15% glycerol stocks at -70 °C before use.

4.1.2 Plant Material

Plant material was provided by Hop Research Institute from Žatec. Female plants of the 'Saaz' hop (*Humulus lupulus* L.) - Osvald's clones 31,72, and 114 were propagated by cuttings and grown in a tissue culture room in sterile conditions at a temperature of 20±2 °C with 16 hours of simulated daylight (Fig. 12). Plants were grown in Magenta boxes and glass tall test tubes with caps in MS-B5 cultivation medium (Fig. 12).

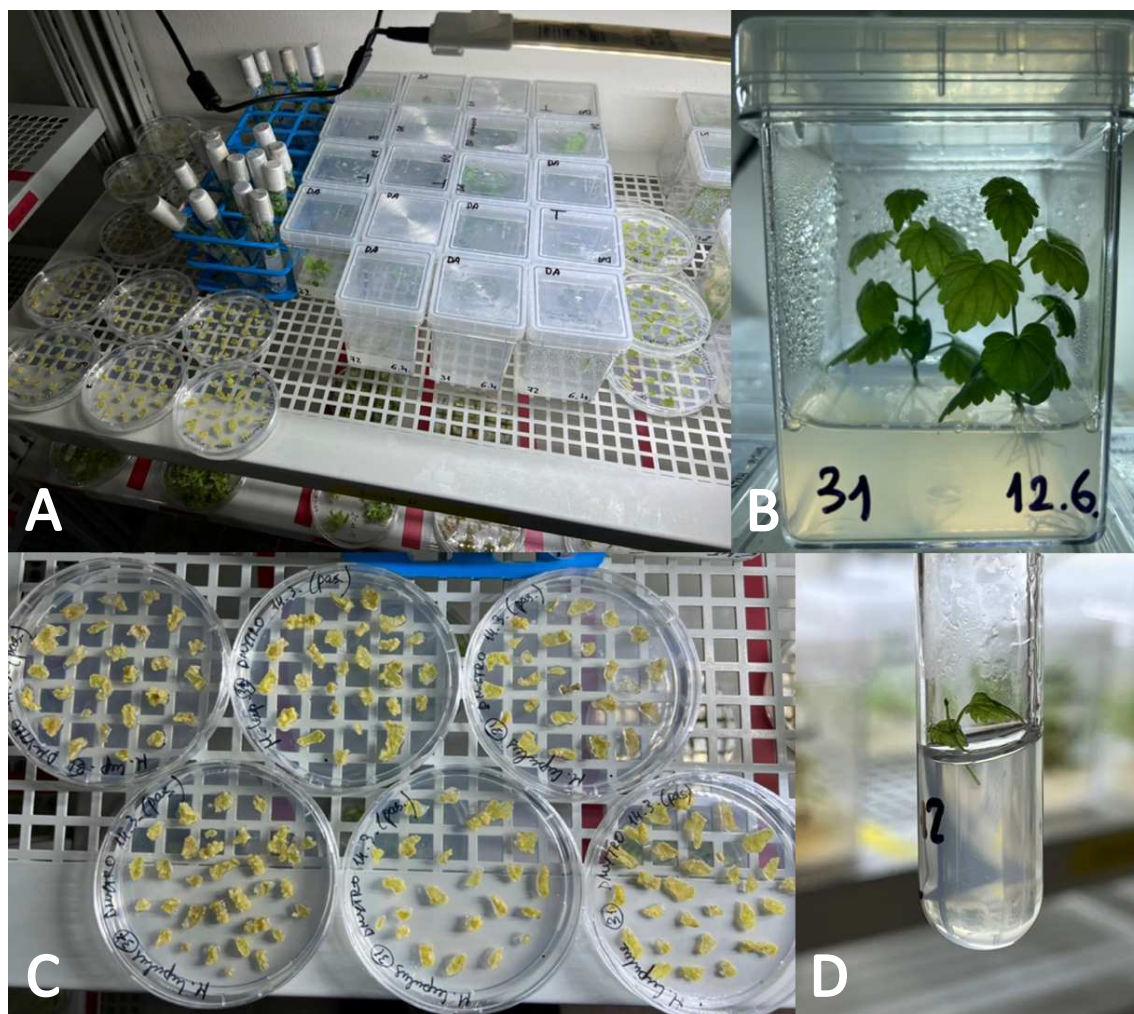


Fig. 12 Photographs showing the arrangement of plants in the cultivation room (A) and types of specialized growing boxes (B), Petri dishes (C), and tubes (D)

4.1.3 Solutions and Media

Hop cultivation media were prepared according to the literature (Horakova et al., 2021; Matoušek et al., 2016; Horlemann et al., 2003). Modified compositions of cultivation media were tested throughout the experiments.

Cultivation medium MS-B5 (1 l) (Horakova et al., 2021):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 6 g agar
- pH 5.6 – 5.8

Resuspension medium (1 l) (Horlemann et al., 2003):

- 4.4 g MS minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml (2.11 mg/l) thidiazuron and 250 µl IAA (0.25 mg/l)
- pH 5.6 – 5.8

Regeneration medium (1 l) (Matoušek et al., 2016):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml zeatin (1 mg/l) and 250 µl IAA (0.25 mg/l)
- 1 ml timentin (400 mg/l) and 30 µl hygromycin (1.5 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Rooting medium (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 500 µl IAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Lysogeny broth – LB (1 l):

- 10 g tryptone
- 5 g yeast extract
- 5 g sodium chloride (NaCl)
- (for solid LB + 15 g agar)
- pH 7.0

GUS staining solution (1 l):

- 250 ml sodium phosphate buffer (200 mM)
- 20 ml EDTA (pH 8) (0.5 M)
- 10 ml ferricyanide [K₃Fe(CN₆)] (100 mM)
- 10 ml ferrocyanide [K₄Fe(CN₆)] (100 mM)
- 10 ml Triton X-100 (10%)
- 25 ml X-Gluc (dissolved in dimethylformamide) (20 mg/ml)
- H₂O
- pH 7.4

Regeneration medium MS-B5 + TDZ + IAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml (2.11 mg/l) thidiazuron and 250 µl IAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Regeneration medium MS-B5 + zeatin + IAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml zeatin (1 mg/l) and 250 µl IAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Regeneration medium MS-B5 + mT + IAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 400 µl mT (4 mg/l) and 250 µl IAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Regeneration medium MS-B5 + TDZ + NAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml (2.11 mg/l) thidiazuron and 250 µl NAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Regeneration medium MS-B5 + zeatin + NAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 1 ml zeatin (1 mg/l) and 250 µl NAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

Regeneration medium MS-B5 + mT + NAA (1 l):

- 4.4 g MS mix of minerals + Gamborg B5 vitamins
- 20 g alpha-D-glucose
- 400 µl mT (4 mg/l) and 250 µl NAA (0.25 mg/l)
- 1 ml timentin (400 mg/l)
- 6 g agar
- pH 5.6 – 5.8

The mixture of MS (Murashige & Skoog medium including Gamborg B5 vitamins) and glucose was dissolved in 1 l of distilled H₂O. The pH was measured and adjusted if necessary by NaOH/KOH or HCl. The medium was then poured into glass bottles and plant agar was added. The final step of the preparation of the **cultivation medium** is sterilization in an autoclave from a certain pressure, the length of sterilization is 20 min at 121 °C.

The **resuspension medium** is the same as the cultivation medium but supplemented with plant growth regulators 1 ml (2.11 mg/l) thidiazuron and 250 µl IAA (0.25 mg/l) and without the addition of plant agar.

The **regeneration medium** consists of a foundation cultivation medium with the addition of plant growth regulators 1 ml zeatin (1 mg/l) and 250 µl IAA (0.25 mg/l) and antibiotics 1 ml timentin (400 mg/l) and 30 µl hygromycin (1.5 mg/l). Antibiotics and plant growth regulators are added in the tissue culture room before spilling the medium into Petri dishes.

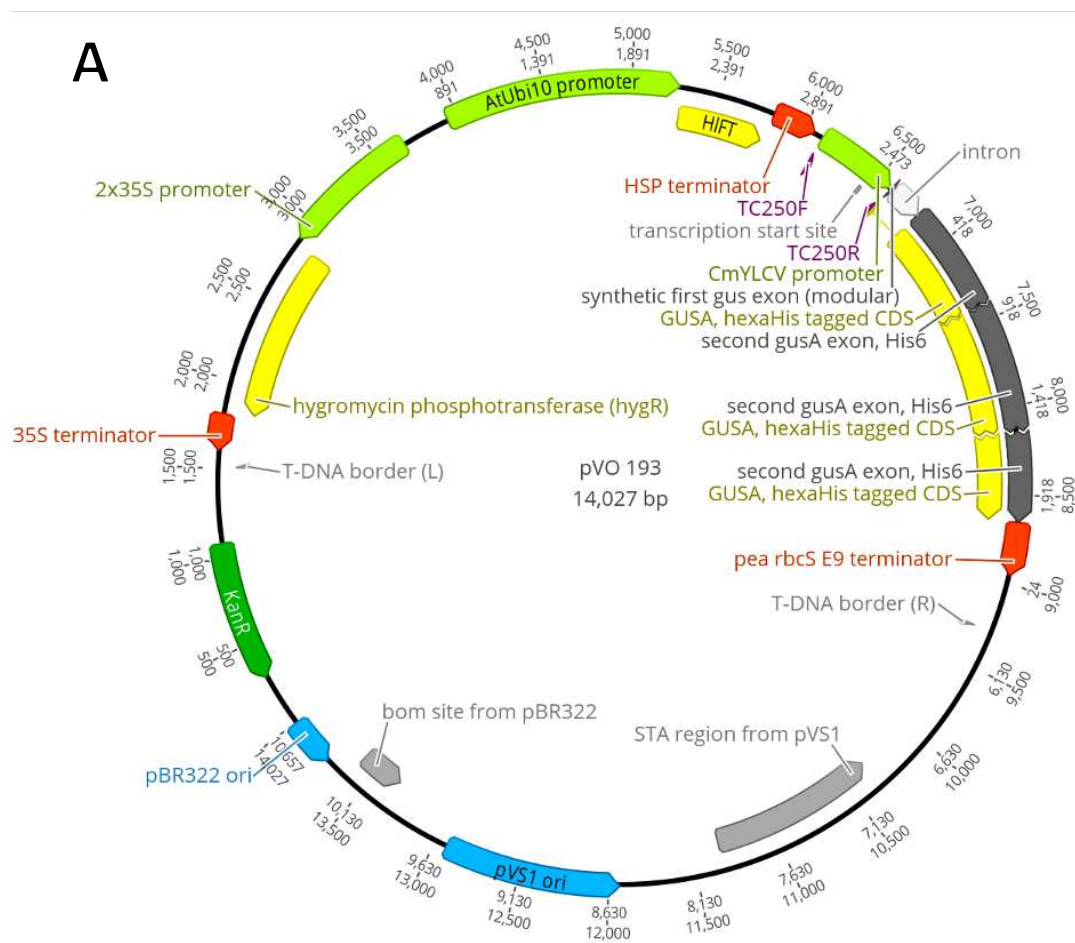
For the preparation of genetic constructs, liquid, and solid LB medium was used at different stages for *E. coli* and *A. tumefaciens* cultivation. Before the explant inoculation liquid LB medium was used for the cultivation of *Agrobacterium tumefaciens* with selection antibiotics and Acetosyringone (activate *Agrobacterium* virulence genes).

4.2 Methods

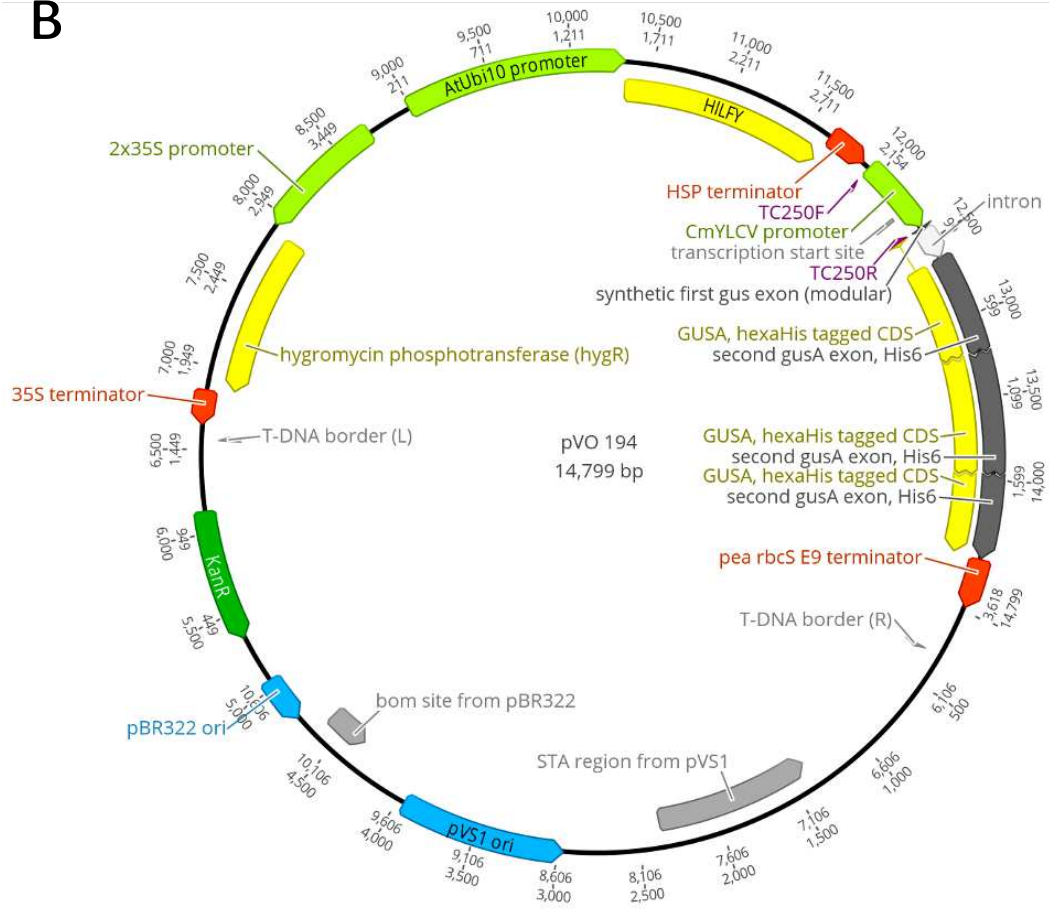
The whole experiment is time-consuming and requires the use of multiple molecular genetic methods. These include RNA and DNA isolation (genomic DNA and plasmid DNA), reverse transcription, blunt-end and golden gate cloning, PCR amplification, and bacterial/plant transformation methods. *In silico* vector assembly and sequence analysis (multiple alignments, primer design) were performed using software tools (ClustalW, MAFFT, Primer3) integrated into *Geneious Prime*.

4.2.1 Design of Vector Constructs

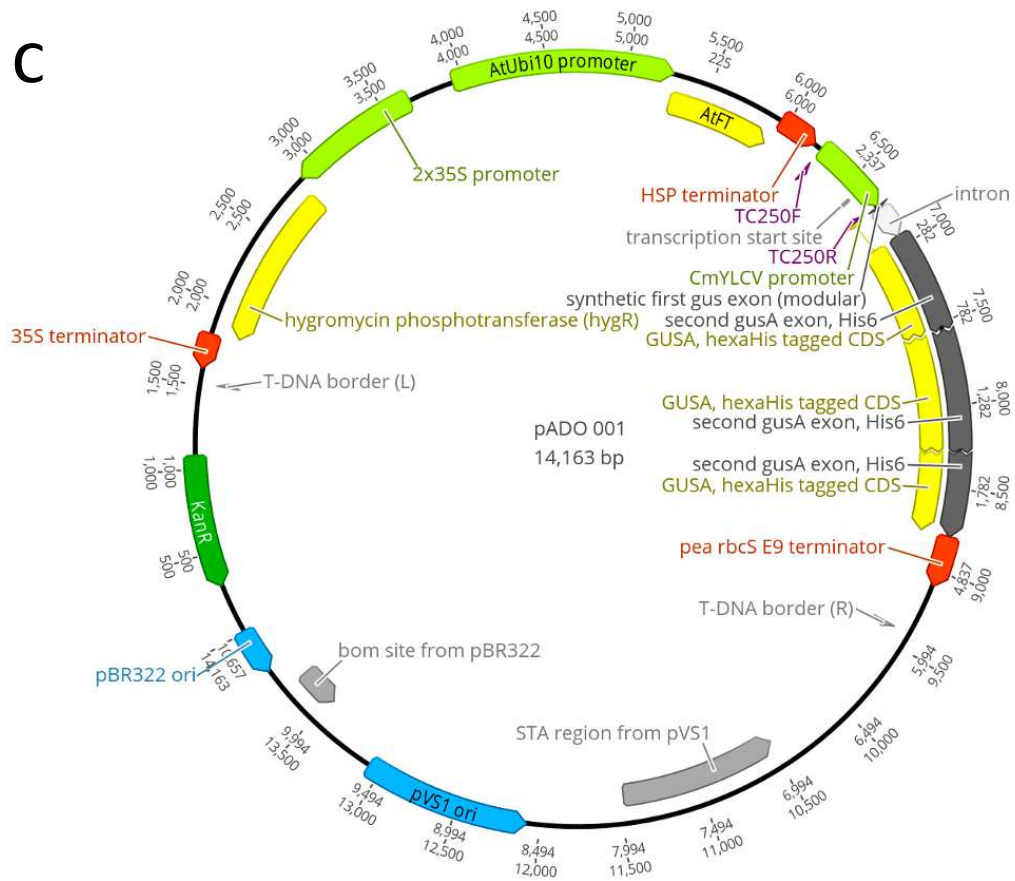
The design of vector constructs for transgenesis was made at the beginning of the experiment. These were constructs pVO193, and pVO194 carrying candidate *HIFT* and *HILFY* genes from hop (*H. lupulus* L.) (Fig. 13). Constructs pADO001 and pADO002 are the same but contain the *AtFT* and *AtLFY* candidate genes from *A. thaliana* (Fig. 13).



B



C



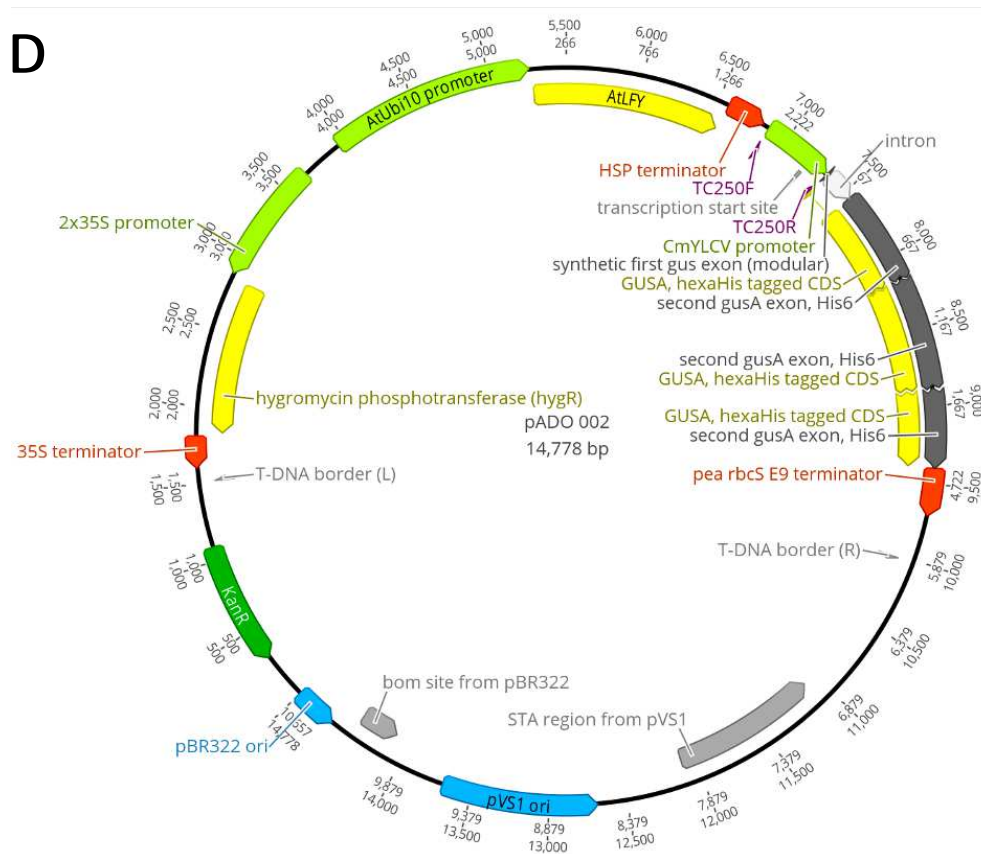


Fig. 13 Schematic representation of genetic constructs for transgenesis. Construct *pVO193* carrying *HIFT* gene (A), *pVO194* carrying *HILFY* (B), *pADO001* carrying *AtFT* gene (C), *pADO002* carrying *AtLFY* gene (D). Designed in Geneious Prime | Bioinformatics Software for Sequence Data Analysis

The binary vector for hop transformation experiments was based on the vector pCambia1301 (Lee & Gelvin, 2008) (Fig. 13). The final binary vector can be divided into T-DNA region and a vector backbone (Fig. 13). The bacterial part contains the *nptII* cassette encoding kanamycin resistance essential for *E. coli* and *A. tumefaciens* selection (Fig. 13). It also contains region (pBR322 ori) facilitates plasmid replication in *E. coli*; the essential for stable plasmid segregation in *Agrobacterium* (pVS1 STA) region; (pVS1 ori) region allowing replication of low-copy plasmids in *Agrobacterium* (Lee & Gelvin, 2008) (Fig. 13). T-DNA region is delimited by TL and TR border 25 base-pair-repeats and contains three expression cassettes: (1) hygromycin phosphotransferase gene encoding plant resistance to hygromycin, (2) gene of interest (*HIFT*, *HILFY*, *AtFT*, *AtLFY*) driven by strong and constitutive AtUbi10 promoter and (3) GUS reporter gene (*uidA*) under the control of CmYLCV promoter (Stavolone et al., 2003) (Fig. 13).

4.2.2 Preparing Vector Constructs

Preparing binary vector constructs for transgenesis is a multi-step process (Fig. 14). The success of each step can be verified using enzymatic cleavage, PCR, bacterial selection, and/or DNA sequencing.

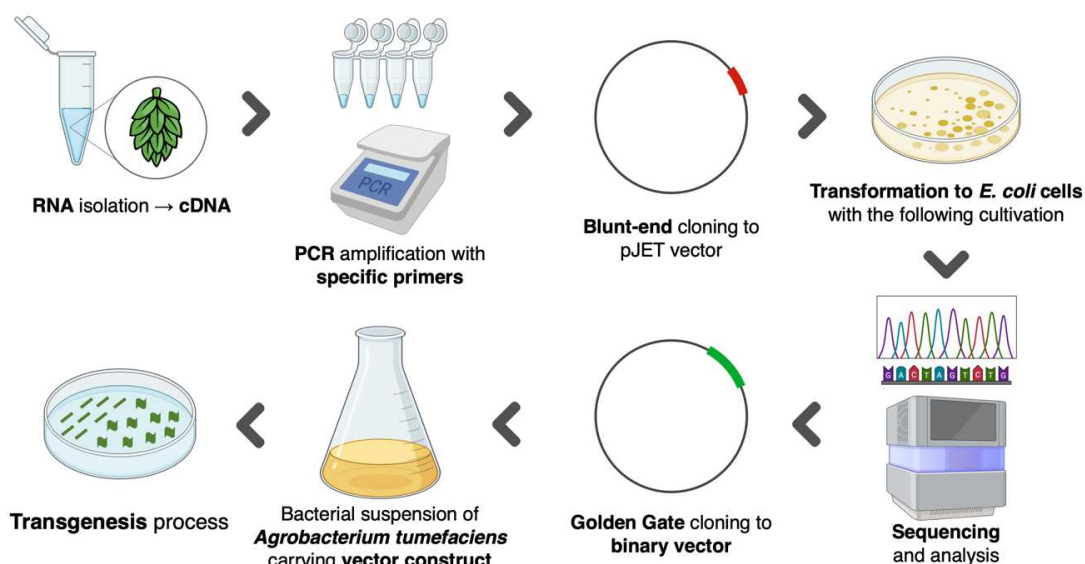


Fig. 14 Scheme of preparation of genetic constructs for transgenesis

Preparation of the construct begins with the **isolation of RNA** by *NucleoSpin RNA Plant Kit* (from Macherey-Nagel) protocol (homogenize samples, lyse cells, filtrate lysate, adjust RNA binding conditions, bind RNA, desalt silica membrane, digest DNA, wash and dry silica membrane, and finally elute DNA) from floral buds and young leaves of *H. lupulus* L./*A. thaliana*. This is followed by **reverse transcription** of the RNA into cDNA by *High-Capacity RNA-to-cDNA Kit* (from Thermo Fisher Scientific). **PCR amplification** of the gene of interest is then performed using specific primers designed for the gene sequence in the *Geneious Prime* software.

Tab. 1 Temperature profile of the PCR reaction

Step	Temperature	Time	Cycle
Initial denaturation	98 °C	3 min	1
Denaturation	98 °C	30 s	39
Annealing	55 °C	30 s	
Elongation	72 °C	3 min	
Final elongation	72 °C	10 min	1

Tab. 2 *Composition of the PCR reaction mixture*

Reagents	Amount per reaction (μl)
H ₂ O	17.4
Q5 5x Phusion buffer	5
dNTPs	0.5
Forward primer	0.5
Reverse primer	0.5
Q5 Phusion polymerase	0.25
Template DNA	1
Total	25.15

Tab. 3 *List of primers*

HIFT_F	5'-ATGCCTAGAGATATCAGGGAC-3'
HIFT_R	5'-ACTGGCTCTGGGGGCAGGAG-3'
HILFY_F	5'-ATGGATCCCGATACCTTCACGG-3'
HILFY_R	5'-CTGCTCATCTCCTTCCCTTCTAA-3'
AtFT_F	5'-ATGCAAATCCGAAACAGTATA-3'
AtFT_R	5'-CGGAGGAAGAAGACTTTAG-3'
AtLFY_F	5'-ATGGATCCTGAAGGTTTCACG-3'
AtLFY_R	5'-GCGACGACTTGCGTTTCTAG-3'

This is followed by **agarose gel electrophoresis** (1.5 % of agarose) and subsequent extraction of PCR products from the gel by the *QIAquick Gel Extraction Kit* (from *QIAGEN*) using the kit protocol. This is followed by **blunt-end cloning** into the pJET1.2 vector.

Tab. 4 *Composition of the ligation reaction*

Reagents	Amount per reaction (μl)
H ₂ O	1.5
PCR product	15
10x T4 DNA buffer	2
pJET vector	0.5
T4 DNA ligase	1
Total	20

The bacterial part starts with **electroporation of *E.coli* cells** using the MicroPulser Electroporator by Bio-Rad (Ec2 program; 0.2 cm cuvettes; voltage = 2.5 kV), followed by plating on Petri dishes with solid LB medium and ampicillin (100 mg/l). Petri dishes

with bacterial suspension were incubated overnight culture at 37 °C. The next step is **plasmid isolation** by *QIAprep Spin Miniprep Kit (from QIAGEN)* protocol. The success of cloning was verified by sequencing (1 µl plasmid; 6.5 µl H₂O, 2.5 µl pJET1.2 Forward Sequencing Primer). **Golden Gate cloning module** is followed by sequencing.

Tab. 5 *Composition of the Golden Gate cloning module reaction*

Reagents	Amount per reaction (µl)
H ₂ O	5.5
plasmid	10
10x T4 DNA buffer	2
pVO164	0.5
BsaI	1
T4 DNA ligase	1
Total	20

This is followed by a **second electroporation of *E.coli* cells** and sequencing according to the principle described above. **Golden Gate cloning into a binary vector** is followed by sequencing.

Tab. 6 *Composition of the Golden Gate cloning into a binary vector*

Reagents	Amount per reaction (µl)
H ₂ O	12
module A plasmid	1
module B plasmid	1
module C plasmid	1
D transformation	0.5
AarI	1
oligonucleotide	0.5
T4 DNA ligase	1
10x T4 DNA buffer	2
Total	20

The final step is to **electroporate the *A. tumefaciens* cells**. This is followed by plating on Petri dishes and culturing for 2-3 days at 22-24 °C. Bacterial colonies were grown in liquid LB medium with selection ATB (kanamycin + gentamicin for GV3101 and kanamycin for EHA105 strains) and then a stock solution was prepared (stored at -70 °C). The finished construct in *A. tumefaciens* cells was used for the **transgenesis** of hop plants.

4.2.3 Transformation of Plant Explants

The precultivation of hop explants takes over 3 days, followed by 2-3 days of cocultivation of explants with *Agrobacterium* (Fig. 15). Then, the explants are transferred to selection regeneration medium for 8-10 weeks (medium is renewed every 2 weeks). Successful regenerants are then transferred to the greenhouse.

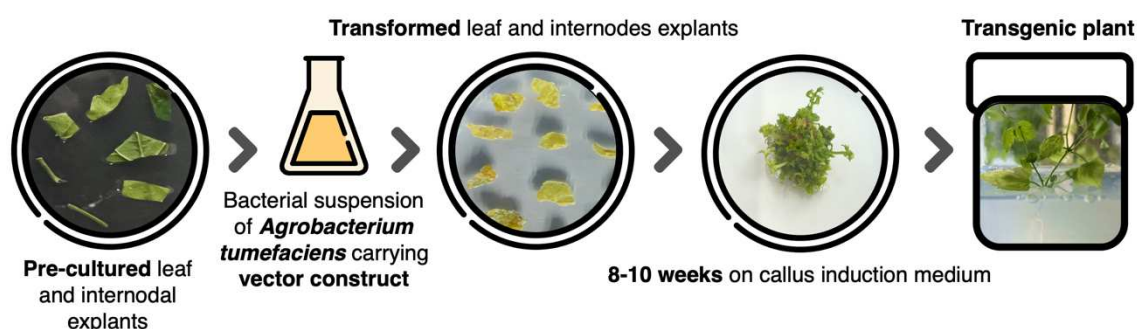


Fig. 15 Schematic view of transformation and regeneration of hop explants

On the first day, explants were prepared from freshly grown hop plants. First, explants were cut in small fragments (5×5 mm leaves and 5-7 mm internodes) and placed on plates containing MS-B5 culture medium for 2 days.

On the second day, the *Agrobacterium* liquid culture is started from 50 ml of liquid LB medium + 50 µl of ATB (when using strain GV3101: 50 µl of kanamycin (50 mg/l) and 50 µl of gentamicin (50 mg/l); for EHA105 - 50 µl of kanamycin) + 50 µl of AS (100 mg/l) + aliquot of *Agrobacterium* was cultured overnight in Erlenmeyer flasks at 28 °C (colonies need to be taken using the tip from the glycerol stock solution or a pre-prepared Petri dish from an *Agrobacterium* culture).

On the third day is the transformation. First, the optical density at 600 nm (OD₆₀₀) of the agrobacterial culture was determined using a spectrophotometer and adjusted to OD₆₀₀ = 0.6-0.7. The culture was then centrifuged at 3000 g for 10 minutes at room temperature. Subsequently, the culture was resuspended in a resuspension medium with the addition of 50 µl of AS. The transformation solution was then gently shaken at 28 °C for 1-2 hours.

The inoculation of the precultured explants in *Agrobacterium* solution took 15 minutes in the desiccator under vacuum infiltration. Then the leaves were dried on a sterile paper towel in a laminar box and placed in Petri dishes on filter paper soaked in a small amount of sterile water with AS (1 µl AS per 1 ml H₂O) or placed on MS-B5 + AS cocultivation medium.

For the next 2-3 days, cocultivation is carried out on a cocultivation medium in the dark (ideally at a temperature of ± 24 °C. After 2-3 days, explants were washed with sterile water and timentin (1 μ l TIM per 1 ml H₂O) (left for 15 min, repeat step 2-4 times without adding timentin) and placed on regeneration medium (callus induction) with timentin + selection ATB (hygromycin).

4.2.4 Cultivation and Regeneration of Plant Material

After the transformation of explants and placing them on a cultivation medium with plant growth regulators and antibiotics, they were transferred on the new medium every 14 days. Cultivation was carried out in a tissue culture room at a temperature of 20 ± 2 °C with 16 hours of simulated daylight. For continuous monitoring of reporter gene expression, explants were collected for histochemical GUS staining assay. After successful regeneration of explants and formation of shoots, shoots were transplanted onto rooting medium or transferred to standard MS-B5 cultivation medium without growth regulators in the Magenta box.

4.2.5 GUS Staining Histochemical Assay

The GUS assay was performed using GUS staining solution under vacuum infiltration in the desiccator. The explants were then incubated at 37 °C for 24 hours. After staining the explants were soaked several times in 80% ethanol (with a time interval of 24 hours) to remove chlorophyll.

4.2.6 Design of gRNA for CRISPR/Cas9 Application

The analysis of the candidate genes *SOC1* and *JOINTLESS (SVP-like)* for CRISPR/Cas9, the construction of genetic maps of the genes, and the design of the guide RNA in *Geneious Prime* were carried out. gRNA sequence was designed to complement the target site to the exon region in the DNA-binding MADS domain, typically around 20 nucleotides (G20nGG) in length.

5 RESULTS

During the experiments, female hop plants were used. The constructs for transformation and the conditions for cultivation of the transformed plants were changed. However, due to time and plant regeneration problems, it was not possible to test all the genetic constructs during the experiment. Anyway, the results of this work are promising for further application of the selected methods to hop plants and continuation/optimization of the experiment.

5.1 Hop Transgenesis Experiment

In the transformation experiment, three different hop genotypes (Osvold clones 31, 72, and 114) were used and two different *Agrobacterium* strains were used (GV3101 and EHA105) into which the construct for transgenesis was inserted. There were four different combinations of strain and vector: GV3101 + pVO193; GV3101 + pVO194; EHA105 + pVO193; EHA105 + pVO194. Transient expression of GUS two-day post-inoculation (Tab. 7-10; Fig. 16) and stable expression three-week post-inoculation (Tab. 11-14; Fig. 17) were monitored. The combination of GV3101 + pVO193 worked best, resulting in 7.9 % of explants with stable expression and EHA105 + pVO193 – 7 %, and this was in genotype 31.

Tab. 7 *Transient expression of different hop explants (GV3101 + pVO193). Leaves and internodes fragments were counted together*

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	98	12	12.2
72	90	7	7.8
114	87	2	2.3

Tab. 8 *Transient expression of different hop explants (GV3101 + pVO194). Leaves and internodes fragments were counted together*

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	80	4	5
72	92	4	4.3
114	73	0	0

Tab. 9 Transient expression of different hop explants (*EHA105 + pVO193*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	61	8	13.1
72	70	5	7.1
114	55	3	5.5

Tab. 10 Transient expression of different hop explants (*EHA105 + pVO194*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	85	5	5.9
72	71	5	7
114	62	0	0

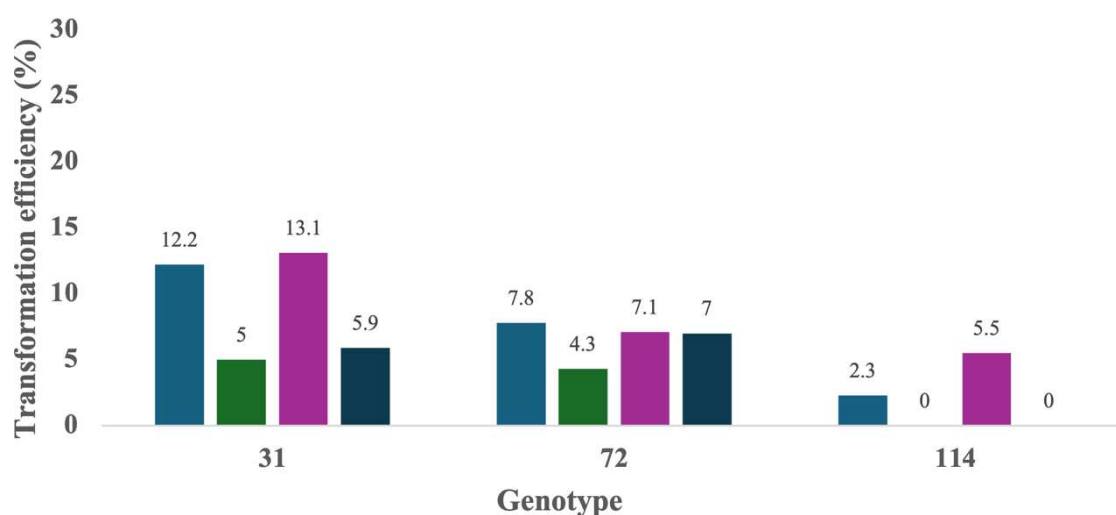


Fig. 16 Graph comparison of *GV3101 + pVO193* (blue color), *GV3101 + pVO194* (green color), *EHA105 + pVO193* (violet color), *EHA105 + pVO194* (dark blue color), **transient expression** of hop explants of different genotypes

Tab. 11 Stable expression of different hop explants (*GV3101 + pVO193*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	76	6	7.9
72	70	2	2.9
114	69	0	0

Tab. 12 Stable expression of different hop explants (*GV3101 + pVO194*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	66	1	1.5
72	69	2	2.9
114	44	0	0

Tab. 13 Stable expression of different hop explants (*EHA105 + pVO193*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	71	5	7
72	67	2	3
114	63	1	1.6

Tab. 14 Stable expression of different hop explants (*EHA105 + pVO194*). Leaves and internodes fragments were counted together

Genotype	Explants	GUS positive	Transformation efficiency (%)
31	45	1	2.2
72	38	3	7.9
114	60	0	0

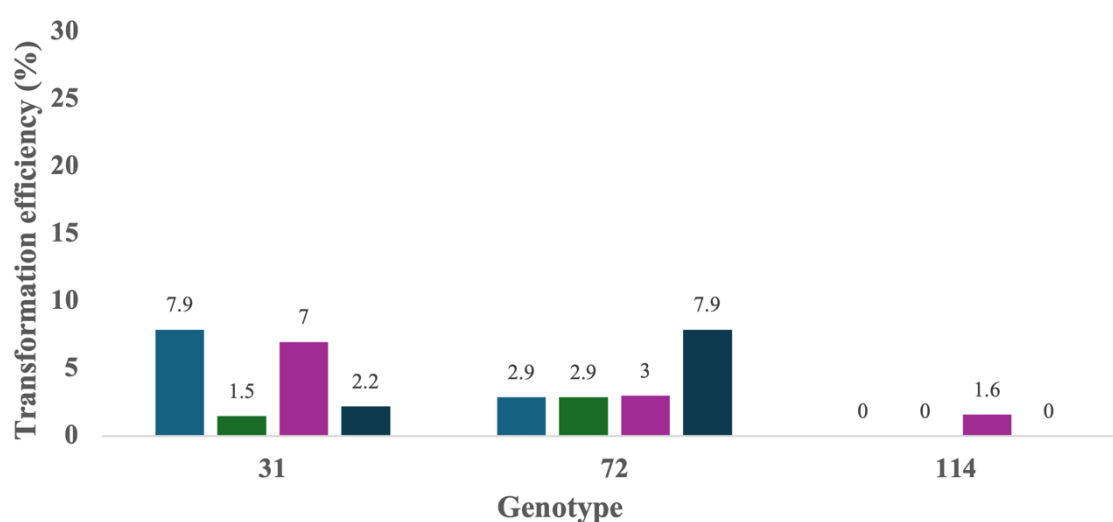


Fig. 17 Graph comparison of *GV3101 + pVO193* (blue color), *GV3101 + pVO194* (green color), *EHA105 + pVO193* (violet color), *EHA105 + pVO194* (dark blue color), **stable expression** of hop explants of different genotypes

Explants were counted together from all Petri dishes in one transformation experiment (separately for each genotype – 31, 72, 114) and the transformation efficiency was estimated, especially for transient GUS expression (the indigo color is spread over the whole explant area) and stable GUS expression (the indigo color is concentrated only at the one place - such foci are the basis of a potential positive transgenic regenerants).

During the whole experiment, which lasted nearly 10 weeks, the GUS-positive explants were photographed to record the success of the transformation. The explants were selected for a visual representation of the results, namely explants with the vector construct GV3101 + pVO193, carrying the *HlFT* gene (Fig. 18-21).

Explants were grown on a selective regeneration medium (MS-B5 + zeatin + IAA + timentin + hygromycin) (Matoušek et al., 2016) (Fig. 18-20). Subsequently, they were transferred to the rooting medium (MS-B5 + IAA + timentin) (Fig. 21).

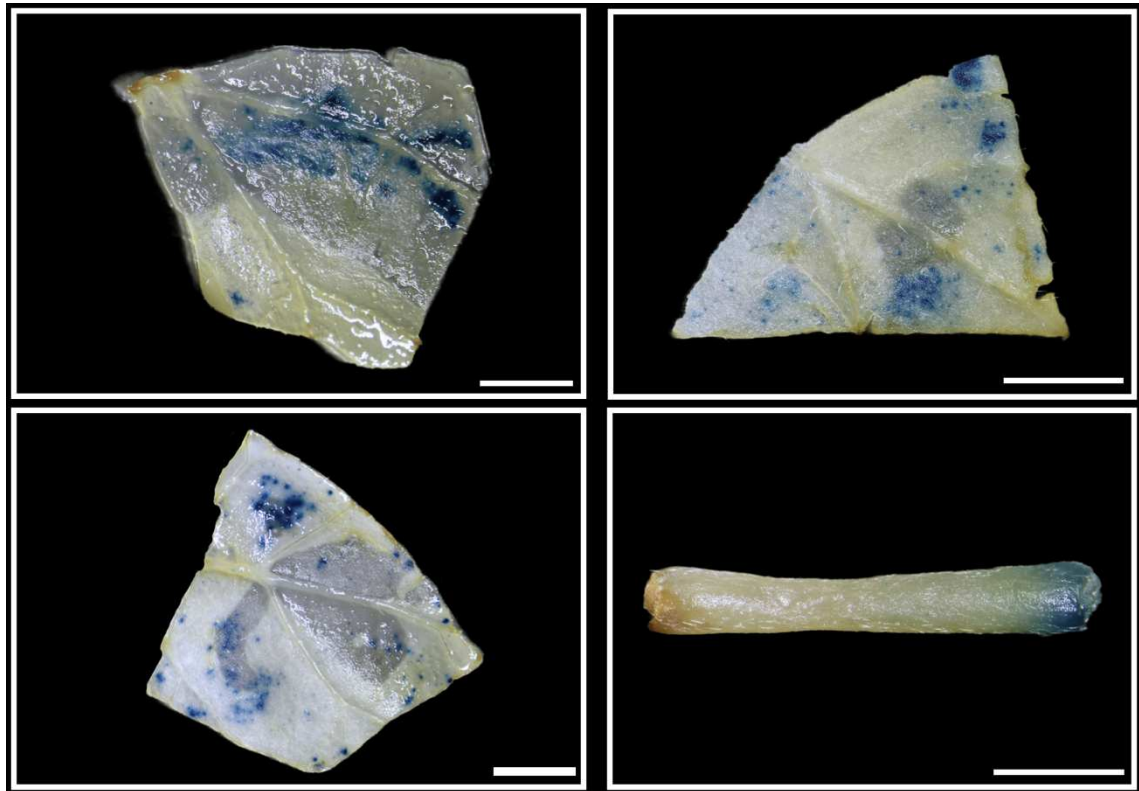


Fig. 18 Two-day post-inoculation hop explants following transformation with a vector construct (pVO193) containing the FT gene. After histochemical GUS staining, they show transient GUS expression (scale = 2000 μ m)

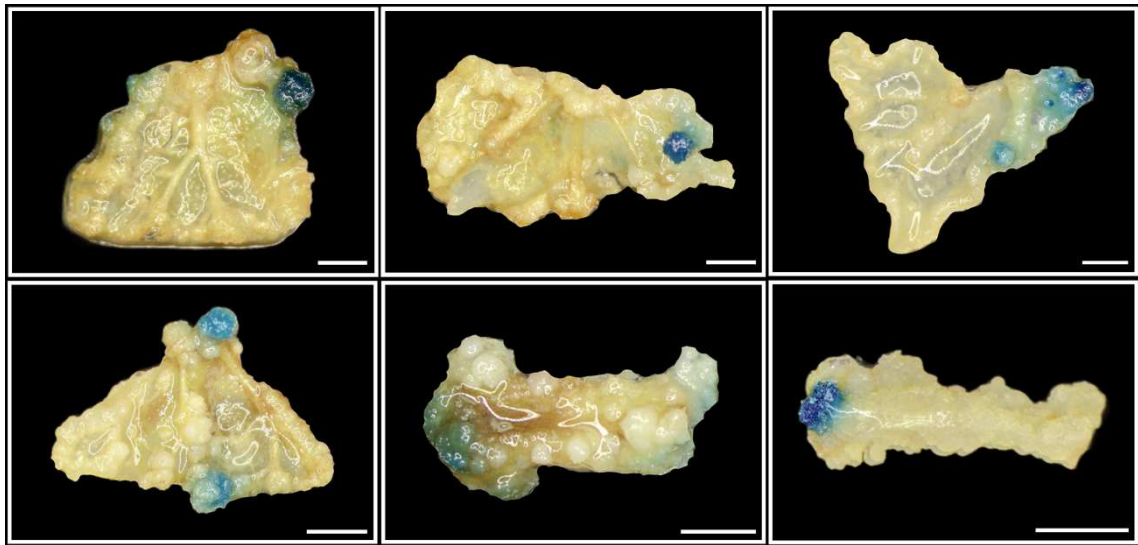


Fig. 19 Three-week post-inoculation hop explants following transformation with a vector construct (pVO193) containing the FT gene. After histochemical GUS staining, they show stable GUS expression (scale = 2000 μ m)



Fig. 20 Five-week post-inoculation hop explants following transformation with a vector construct (pVO193) containing the FT gene showing signs of regeneration on selective medium with hygromycin (scale = 2000 μ m)

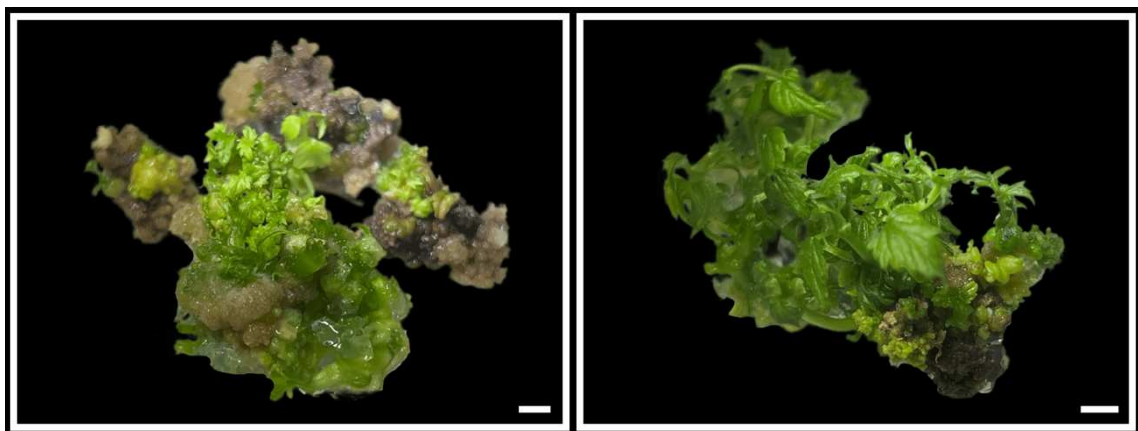


Fig. 21 Ten-week post-inoculation hop explants following transformation with a vector construct (pVO193) containing the FT gene showing appreciable regeneration on rooting medium (scale = 2000 μ m)

5.2 Growth Regulator Test

In this experiment, was tested the *in vitro* regeneration frequency of transformed hop explants. Three genotypes, 31, 72, and 114, were used. Testing was carried out on six different media and combinations of plant growth regulators were created: MS-B5 + TDZ + IAA, MS-B5 + zeatin + IAA, MS-B5 + mT + IAA, MS-B5 + TDZ + NAA, MS-B5 + zeatin + NAA, MS-B5 + mT + NAA (Tab. 15-20; Fig. 22). Each medium also contained timentin but no hygromycin added.

As a result, the percentage of successful explants was obtained and then an evaluation of the best type of medium for regeneration was made MS-B5 + zeatin + IAA.

Tab. 15 Internodes fragments regeneration on MS-B5 + TDZ + IAA medium

Genotype	MS-B5 + TDZ + IAA (internodes)	MS-B5 + TDZ + IAA (regenerated)	Regeneration efficiency (%)
31	44	10	22.7
72	29	7	24.1
114	27	1	3.7

Tab. 16 Internodes fragments regeneration on MS-B5 + zeatin + IAA medium

Genotype	MS-B5 + zeatin + IAA (internodes)	MS-B5 + zeatin + IAA (regenerated)	Regeneration efficiency (%)
31	41	24	58.5
72	40	18	45
114	31	5	16.1

Tab. 17 Internodes fragments regeneration on MS-B5 + mT + IAA medium

Genotype	MS-B5 + mT + IAA (internodes)	MS-B5 + mT + IAA (regenerated)	Regeneration efficiency (%)
31	30	4	13.3
72	41	11	26.8
114	19	3	15.8

Tab. 18 Internodes fragments regeneration on MS-B5 + TDZ + NAA medium

Genotype	MS-B5 + TDZ + NAA (internodes)	MS-B5 + TDZ + NAA (regenerated)	Regeneration efficiency (%)
31	44	5	11.4
72	43	2	4.65
114	47	0	0

Tab. 19 Internodes fragments regeneration on MS-B5 + zeatin + NAA medium

Genotype	MS-B5 + zeatin + NAA (internodes)	MS-B5 + zeatin + NAA (regenerated)	Regeneration efficiency (%)
31	42	8	19
72	55	4	7.2
114	38	0	0

Tab. 20 Internodes fragments regeneration on MS-B5 + mT + NAA medium

Genotype	MS-B5 + mT + NAA (internodes)	MS-B5 + mT + NAA (regenerated)	Regeneration efficiency (%)
31	22	2	9.1
72	13	2	15.4
114	20	1	5

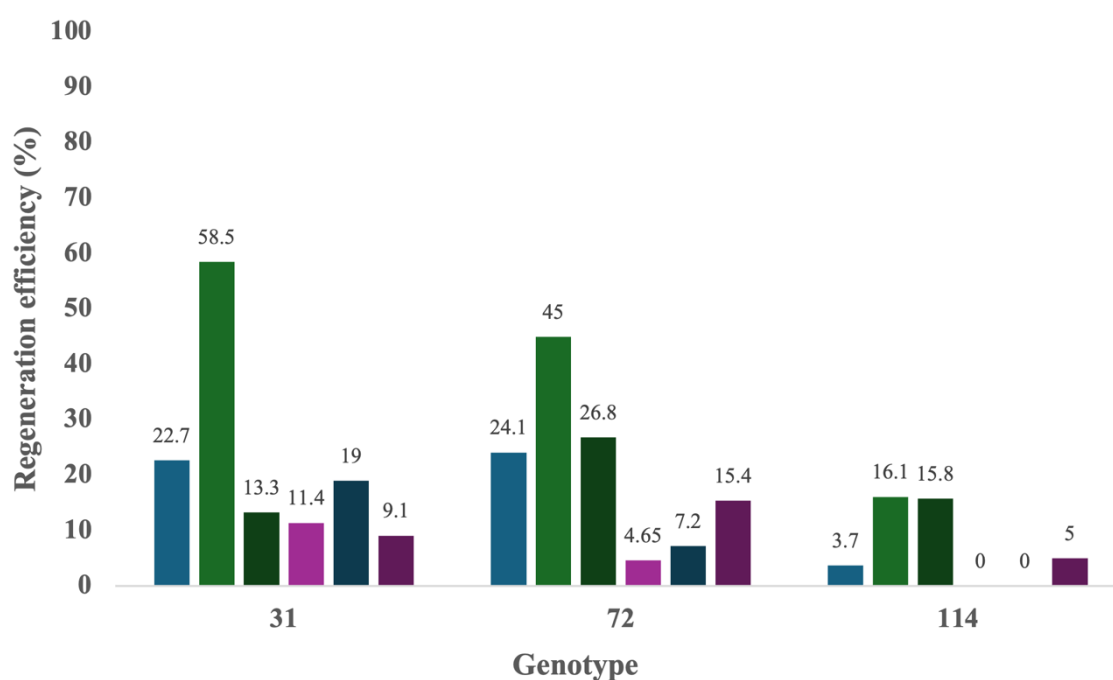


Fig. 22 Graph comparison of MS-B5 + TDZ + IAA medium (blue color), MS-B5 + zeatin + IAA medium (light green color), MS-B5 + mT + IAA (dark green color), MS-B5 + TDZ + NAA medium (light violet color), MS-B5 + zeatin + NAA medium (dark blue color), MS-B5 + mT + NAA (dark violet color) regeneration of hop internodes of different genotypes

5.3 Design of the CRISPR/Cas9 gRNAs

The creation of CRISPR/Cas9 genomic constructs begins with the selection of regions in the genome that need to be changed. This is usually done with the help of specialized software tools (*Geneious Prime*) that help identify optimal sites for precise modification.

This design includes the RNA molecule that will guide Cas9 to the target site, as well as the DNA sequence that will be inserted or altered. Two ideal gRNA1 (GAATAATAATCTAATACAGG) and gRNA2 (GAGACTGAACTATTCATTGG) sites on the *SOC1* gene and two gRNA1 (GTACTGTCGTAATCTTGGGG) and gRNA2 (GTATGTGATCACTGAAGAGG) sites on the *JOINTLESS (SVP-like)* gene were found and selected. *SOC1* and *JOINTLESS (SVP-like)* genes are derived from hop. A genetic map was constructed to visualize the experiment (Fig. 23).

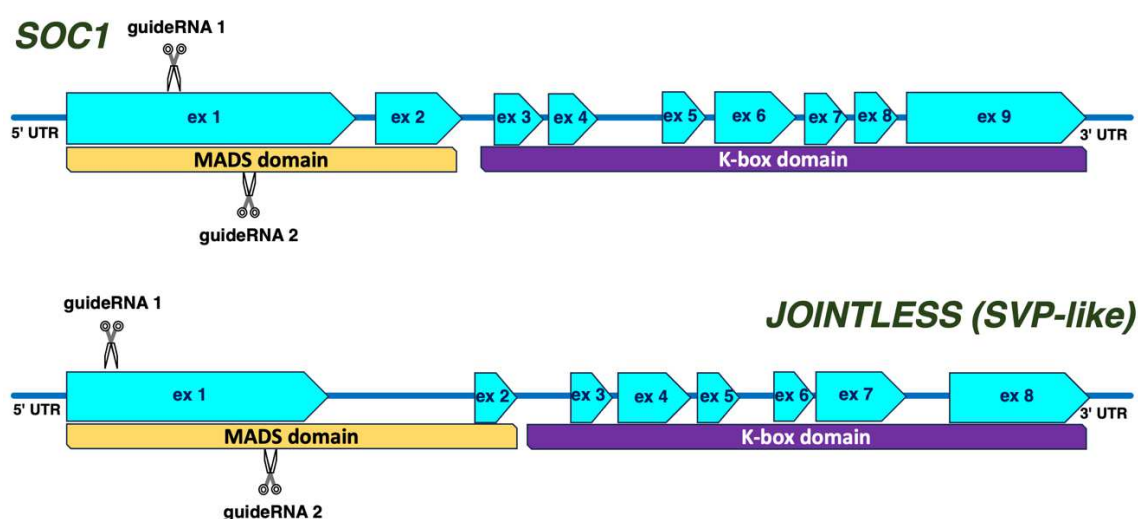


Fig. 23 Schematic view of CRISPR/Cas9 design. Design guide RNA to target genes

The important point is that the fragments must be located in the exon region and simultaneously in the MADS DNA-binding domain for the subsequent knockout to be successful.

6 DISCUSSION

The genetic transformation of crops is a highly relevant and debated topic. An unprecedented number of plant genome modification experiments have been carried out all over the world, not only on model plants (Lowe et al., 2016). The main goal of current research in the field of plant biotechnology is to improve the diverse properties of plants. This effort has led to the development of a wide spectrum of molecular genetic and transformation techniques. These methodological advances allow for the investigation and faster generation of desired phenotypes. An important field is the investigation of the genomes of non-model plant species and the development of approaches to improve their manipulation.

For the research in this thesis, the hop plants (*Humulus lupulus* L.) variety 'Saaz' - Osvald's clones 31,72, and 114 (Nesvadba et al., 2020) were selected as the experimental material. Although many hop varieties are distributed worldwide, Saaz represents an important cultivar in terms of its long tradition in the Czech Republic. The genetic transformation of hop opens new possibilities in the field of improvement of this plant for various applications such as improved abiotic and biotic stress resistance, modified metabolite production, and wider genetic diversity. On the other hand, the main issue is low transformation efficiency.

This work builds on hop genetic transformation experiments performed at the Laboratory of Plant Developmental Genetics, Institute of Biophysics of the Czech Academy of Sciences (Horakova et al., 2021). The research aim was to perform the transformation of hop and obtain transgenic material with the subsequent creation of experimental hop lines with accelerated onset of flowering. Such material will be suitable for further functional analyses in the future. This is also due to the complexity of the hop as a model organism (long vegetative phase, plant height, genome size, and complexity).

The accelerated onset of flowering can be experimentally induced by increasing the expression of genes involved in the transition from the vegetative to the generative phase. This has been successfully done in woody plants (Mohamed et al., 2010). Various genetic constructs have been designed to overexpress the *FT* or *LFY* gene, which participates in the initiation and regulation of flowering (Klocko et al., 2016; Siriwardana & Lamb, 2012). Most of the successful transformation experiments in this work were performed using a vector carrying the *FT* gene from hop. The sequence of the *FT* gene from hop was analyzed and its exact variant, *HIFT3*, was determined and used in the experiment. This

is important to know in advance because of the different expression conditions of the different variants of this gene (Gutiérrez et al., 2023). The *LFY* gene was chosen as an alternative variant for this study because of its function and its involvement in the mechanism of floral development.

The *Agrobacterium*-mediated transformation method was chosen for hop transgenesis. Two different strains of *Agrobacterium tumefaciens* were tested - GV3101 with very high virulence (Chetty et al., 2013; Lian et al., 2022) and EHA105. The EHA105 strain was selected as similar to the EHA101 strain, which has already been used for the transformation of hop (Horlemann et al., 2003) and several other plant species (Hiei et al., 1994; Frame et al., 2002). It is very important to maintain a certain temperature in experiments with *Agrobacterium*, not only during pre-cultivation of the transformation solution but also during cocultivation with plant material. Temperature plays an important role and the goldilocks for the coculture of explants with *Agrobacterium* is 24 ± 2 °C (Fullner & Nester, 1996). For the cultivation of *Agrobacterium* culture alone, the temperature can reach 28 °C because the T-DNA transfer machinery is known to work properly only at a temperature range of 19-28 °C (Fullner & Nester, 1996; Uranbey et al., 2005). Also, a very important factor is a very sharp cutting tool to avoid tissue maceration and to create a precise cutting surface, which also increases the efficiency of the transformation.

When the experiment was attempted without pre-cultivation of the hop explants, no successful results were obtained. This may be related to the ability of acclimation of the plant tissues to the surrounding conditions. The percentage of survival and regeneration was almost zero. However, the pre-cultured explants were able to obtain explants with both transient and stable expressions. Both leaves and internodes were used and counted together. However, leaf explants were more often used for GUS histochemical staining than for regeneration. The combination of GV3101 + pVO193 worked best, resulting in 7.9 % of explants with stable expression and EHA105 + pVO193 – 7 %, and this was in genotype 31, which is consistent with an earlier study (Horakova et al., 2021). The other lines 72 and 114 showed lower regenerative capacity. The cause of the lower expression was not found in comparison with a similar experiment (Horakova et al., 2021). A binary vector construct with the *LFY* gene (pVO194) was also used for transformation but a lower percentage of GUS-positive explants was achieved. It can be assumed that the problem is at the level of the original plant material.

The construction of transgenic plants is one of the prerequisites for the successful application of modern genetic engineering tools - increasing the efficiency of transgenesis is therefore another important point for the analysis of gene function in hop. Currently, the transformation efficiency of the hop is very low and depends a lot on the conditions of cultivation both during cocultivation with *Agrobacterium* and after transformation on regeneration medium. Therefore, it is important to carefully consider the selection of the specific form of plant growth regulator and its concentration during the cultivation of explants to achieve the desired physiological and morphological development of the plants.

Growth regulator assessment experiments were conducted to improve *in vitro* regeneration of hop plants. Various plant growth regulators may have different effects on explants cultured on a medium containing that regulators. These differences may be due to the chemical structure of the growth regulator, conjugate formation, its concentration, signal transduction, abundance in the plant tissue, and also interactions with other components of the medium. Media with such growth regulators as zeatin, thidiazuron, meta-topolin, indole-3-acetic acid, and 1-naphthaleneacetic acid were used. The medium MS-B5 + zeatin + IAA and MS-B5 + TDZ + IAA were shown to work best, however, the best number of successful regenerants was achieved on the medium MS-B5 + zeatin + IAA. The effectiveness of this medium is also confirmed in other studies (Matoušek et al., 2016; Horakova et al., 2021). For this experiment, only internodes were used because they have better regeneration capacity (Austrian & Matoušek, 1994) and it was performed on transformed explants but without the selection of antibiotic hygromycin in the regeneration medium to observe only the regeneration capacity of explants on different types of media without additional inhibition of growth of GUS negative plants.

To increase the transformation efficiency at later stages, vector constructs containing *Arabidopsis thaliana* *FT* and *LFY* genes were created. Although they have not been tested, these constructs will be used for further studies in this area.

A successful application of the CRISPR/Cas9 system for targeted mutagenesis of hop has been described some time ago (Awasthi et al., 2021; Maher et al., 2020). To obtain mutant material in the future, vectors with CRISPR/Cas9 nucleases would be constructed, which can be easily and quickly created in different variations (different versions of Cas9 proteins, promoter/terminator sequences) with several gRNAs for the gene of interest due to the modular way of cloning.

In this work, gRNAs for the *JOINTLESS* (*SVP-like*) and *SOC1* genes, which are involved in the regulation of flowering, were designed. Knockout of these genes should result in the desired phenotype (faster onset of flowering). However, it was not possible to fully implement this experiment due to a lack of plant material in the final stage of the thesis.

Even though a high percentage of transformation success was not achieved, this work is beneficial as it provides information on transgenesis and editing of the hop genome. From the above-mentioned facts, it is clear that the regeneration of hop explants is highly dependent on cultivation circumstances and the process of its transformation is very challenging and problematic but provides a wide scope for further analysis and opportunities for applied research, which is a promise for future improvement of hop transgenesis.

7 CONCLUSIONS

In an era characterized by a growing global population and environmental concerns, ensuring sustainable agriculture is imperative. As the world faces the need to produce more food with fewer resources, plant biotechnology is emerging as a critical tool to address these pressing challenges. By exploring genetic transformation methods, we have seen how advances in *Agrobacterium*-mediated transformation, biolistic gene delivery, and other innovative techniques have enabled the introduction of foreign genes into plant genomes. The ability to introduce useful genes into plants and to precisely modify their genomes promises improved yields, pest and disease resistance, and the development of new agricultural solutions.

In the pursuit of sustainable agriculture and the ever-pressing need to feed a growing global population, the field of plant biotechnology has emerged as a beacon of hope. This thesis has delved into the realm of modern methods for transgene delivery and genome editing in plants, describing the transformative potential of these technologies for revolutionizing agriculture as we know it.

In closing, the journey through modern methods for transgene delivery and genome editing in plants has revealed a landscape of unprecedented potential. These technologies offer us the possibilities to address critical agricultural challenges, from crop resilience to enhanced nutrition, all while urging us to approach this transformative power with wisdom and care. As we look to the future, the fusion of science, ethics, and innovation holds the key to a more sustainable and nourished world.

8 LIST OF ABBREVIATIONS

<i>AGL24</i>	<i>AGAMOUS-LIKE24</i> gene
ATB	antibiotic
<i>A. tumefaciens</i>	<i>Agrobacterium tumefaciens</i>
<i>API</i>	<i>APETALA1</i> gene
AS	acetosyringone
<i>A. thaliana</i>	<i>Arabidopsis thaliana</i>
<i>bar</i>	<i>bialaphos resistance</i> gene
ChIP	chromatin immunoprecipitation
ChIP-Seq	Chromatin Immunoprecipitation Sequencing
CMV	<i>Cauliflower mosaic virus</i>
<i>COI</i>	<i>CONSTANS1</i> gene
CRISPR	clustered regularly interspaced short palindromic repeats
Cas9	CRISPR-associated protein 9
Cas13	CRISPR-associated protein 13
dCas9	Cas9 endonuclease dead/dead Cas9
DDA	data-dependent acquisition
DIA	data-independent acquisition
DNA	deoxyribonucleic acid
cDNA	complementary DNA
T-DNA	transfer DNA
<i>E. coli</i>	<i>Escherichia coli</i>
<i>FLC</i>	<i>FLOWERING LOCUS C</i> gene
<i>FT</i>	<i>FLOWERING LOCUS T</i> gene
<i>AtFT</i>	<i>Arabidopsis thaliana FLOWERING LOCUS T</i> gene
<i>HlFT</i>	<i>Humulus lupulus FLOWERING LOCUS T</i> gene
GFP	green fluorescent protein
GMO	genetically modified organism
<i>gox</i>	<i>glyphosate oxidoreductase</i> gene
GUS	β-glucuronidase
HDR	homology-directed repair
<i>H. lupulus</i>	<i>Humulus lupulus</i>
<i>hpt</i>	<i>hygromycin phosphotransferase</i> gene
HR	herbicide resistant
HSP	heat shock protein
H ₂ O	water
IAA	indole-3-acetic acid
LB	lysogeny broth
<i>LFY</i>	<i>LEAFY</i> gene
<i>AtLFY</i>	<i>Arabidopsis thaliana LEAFY</i> gene
<i>HlLFY</i>	<i>Humulus lupulus LEAFY</i> gene
MADS	MADS-box gene family

MOAC	metal oxide affinity chromatography
MS	Murashige & Skoog
mT	meta-topolin
NAA	1-naphthaleneacetic acid
NHEJ	non-homologous end joining
<i>nptII</i>	<i>neomycin phosphotransferase</i> gene
OD	optical density
PCR	polymerase chain reaction
qPCR	quantitative polymerase chain reaction
<i>PDS</i>	<i>phytoene desaturase</i> gene
<i>HIPDS</i>	<i>Humulus lupulus PDS</i> gene
PEG	polyethylene glycol
PRSV	<i>Papaya ringspot virus</i>
PTM	post-translational modification
RNA	ribonucleic acid
gRNA	guide RNA
sgRNA	single guide RNA
RNA-Seq	RNA sequencing
scRNA-Seq	single-cell RNA sequencing
RNP	ribonucleoprotein
<i>SOC1</i>	<i>suppressor of overexpression of CO1</i> gene
SV	<i>Simian virus</i>
TALEN	transcription activator-like effector nuclease
TDZ	thidiazuron
TFBS	transcription factor binding site
TIM	timentin
TMV	<i>Tobacco mosaic virus</i>
ZFN	zinc finger nuclease

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Fig. 18 Two-day post-inoculation hop explants following transformation with a vector construct (pVO193) containing the *FT* gene. After histochemical GUS staining, they show transient GUS expression (scale = 2000 μ m)

Fig. 19 Three-week post-inoculation hop explants following transformation with a vector construct (pVO193) containing the *FT* gene. After histochemical GUS staining, they show stable GUS expression (scale = 2000 μ m)

Fig. 20 Five-week post-inoculation hop explants following transformation with a vector construct (pVO193) containing the *FT* gene showing signs of regeneration on selective medium with hygromycin (scale = 2000 μ m)

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APPENDICES